

MAJOR PROJECT

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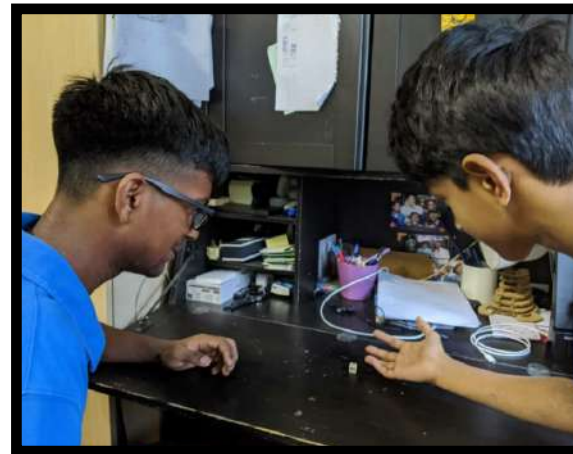
Consideration of Design Need

The Situation

The 2 Bharadwaj brothers, Satvik and Varun, love to **bet** with each other after a long day of school and homework. When both of them find themselves **bored** with no work on their hands, they eventually meet each other in the older brother, Satvik's bedroom to take shots to profit off each other. After finding a temporary suitable device/object and making multiple bets, it always appears that **one person has a great advantage over the other**. The losing brother then blames the **fault** on the **gadget** and asks for their money in return. This **leads to arguments** which always resolves with each player getting back their original amount and the gadget being thrown away or unused. In other cases, the betting device **isn't entertaining enough** and the brothers' aren't having fun. Satvik is currently in grade 9 (15 years old) and Varun is currently in grade 5 (11 years old).



Satvik's bedroom



A photo of the Bharadwaj brothers betting with a dice

The Problem

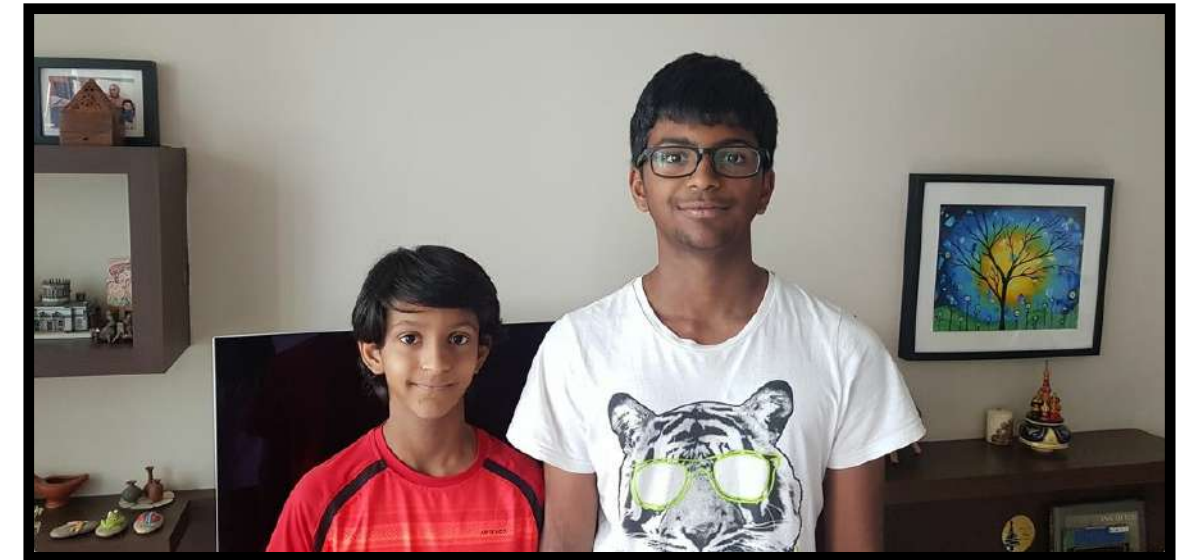
Satvik and Varun can both **never decide on a suitable gadget or object** to place bets on. In the past, they have used dices, multiple online random number generators and even coins. All of which have been felt to be rigged or boring by at least one of the brothers.

The Consequences of the Problem

After an argument on a **rigged game**, the 2 brothers end up **not talking to each other** for at least half a day and putting them both in a **bad mood**. This may occasionally **lead to some violence** and **more arguments** and **family discussions**.

Design Brief

I propose to design a product that **solves the Bharadwaj brothers' problem**. The product will have to be an **electronic betting tool** that is **light, portable, entertaining** and have **many outcomes** so that there can be **multiple forms of betting with one device**. This will satisfy both the brother's needs and minimise arguments between them regarding this issue. The product will have to be completed before March 2022 and should be below or around \$80.



Clients

Name	Varun Bharadwaj	Satvik Bharadwaj
Age	11	15
Grade	Grade 5	Grade 9
Nationality	Indian	Indian

The Clients' Needs

After having a discussion with both clients, they have both **agreed** that they would use an **electronic betting tool** if I could provide one for them. The electronic betting tool needs to have a **high number of possible outcomes** compared to their previous betting gadgets for more **entertainment** and will also then definitely agreed to be fair. However, they have requested a few **specifications and requirements** for the betting tool to suit their needs. They have specified that the gadget needs to be **light** and **portable** and an extra perk would be to have **sound**. The betting tool will also have to be **durable** and **last over many uses** as the brothers will use the device many times.

Light and portable	Produces sound
Many possible outcomes	Can be used in multiple ways (multiple games)
Takes up minimal space	Durable

Product Analysis

C2

Purpose of product analysis

The purpose of analysing multiple products is to take a look at the **strengths** and **weaknesses** of different **existing products** and to compare multiple **factors** when deciding what to take **inspiration** for my product. This will help me put in place a few **specifications** for the product.



Product 1 - Saturn auto roulette wheel

<https://bit.ly/3vqshDF>

Aesthetics: Modern, flashy

Cost: \$2300 SGD

Customer: For people who would like to spend a lot of money on a roulette wheel

Ergonomics: Large, not too hard to use, three modes, automatic, driven and manual

Function: Rolls a random digit from 1 to 36

Sustainability: Uses a lot of plastic and many electronics

Safety: Large and heavy, user must handle with care



Product 3 - eDice

<https://bit.ly/2PZT3CJ>

Aesthetics: Modern, flashy, small

Cost: \$50 SGD

Customer: Made for someone who needs a cool dice with many modes

Ergonomics: Very easy to carry, easy to use, many different modes

Function: Electronic dice, may different options and modes to select

Sustainability: Uses a lot of electronics but easily recharged so the eDice can last a long time

Safety: Might get lost



Product 2 - Hardware random number generator

<https://bit.ly/3eDyTsp>

Aesthetics: Simple, dull, straightforward, tall

Cost: \$20 SGD

Customer: Made for indecisive people

Ergonomics: Seems easy to use, only a few buttons and switches, very portable

Function: There are two modes for this product. A yes or no answer, six random digits or two rolls of a six sided dice

Sustainability: Doesn't use much too much materials. powered using batteries that have to be swapped over time

Safety: Client will have to be careful when changing batteries

Pros

Product 1: **Modern** and flashy, has a **wide set of outcomes**, **three different modes** (can be based on skill or luck), can **last long**, comes in **multiple colours**

Product 2: **Cheap**, **minimalistic**, **simple to use**, **very portable**

Product 3: **Small**, **attractive/flashy**, many **different modes**, **portable**, can **last long** after recharged

Cons

Product 1: **Expensive**, takes up a lot of **storage space**, uses a lot of **plastic** and **electronics**

Product 2: **More of a decision maker** than a game tool, customers will get **bored** easily, uses **single use batteries**

Product 3: May be easily **lost**, **battery inefficient** as it can be left on

Evaluation

Overall, the **first product** seems to match my clients' needs the most with **multiple modes** and a **wide set of outcomes**, however, the first product is way too **expensive** and takes up a lot of **space** so it can't be easily **stored** and isn't very **portable**. The other 2 products may not be as great but both have certain **aspects** which can have **inspiration** taken from.

Collection of Relevant Data

Purpose

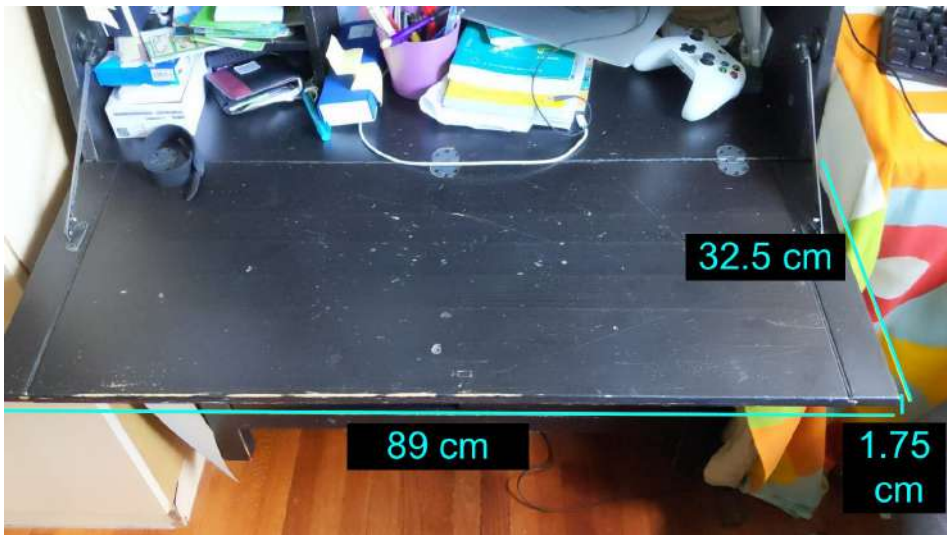
The purpose of collecting relevant data is to understand the **environment** of my product and how objects in its vicinity will **effect** the usage of the product. This will help me to create **specifications** and will allow me to see what **restrictions** I have due to the environment. I will also look at ways to improve the **efficiency** of the product for my client

Measurements

Data of the desk

Length: 32.5 cm
Width: 89 cm
Thickness: 1.75 cm
Area at the top: 2892.5 cm squared
Material: Wood, MDF

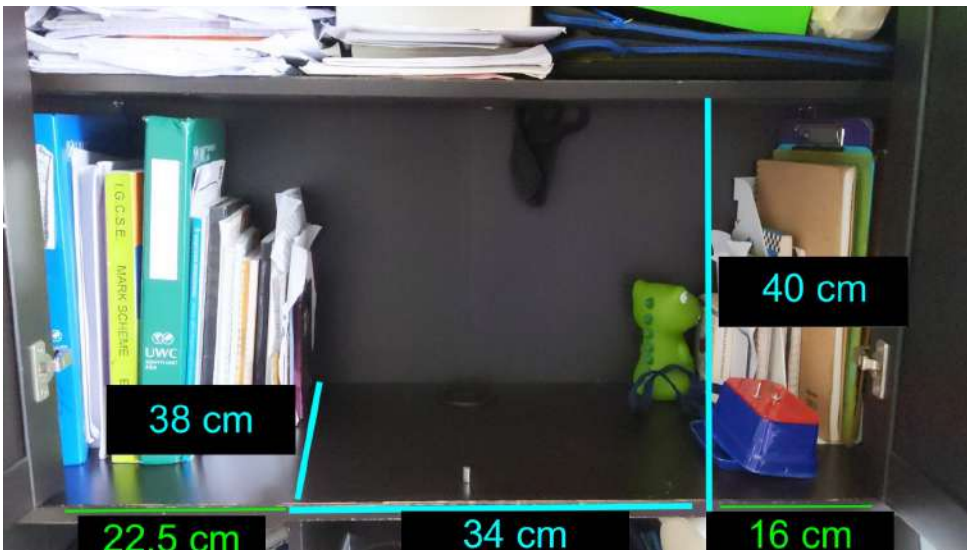
Since the desk has a decently large surface area on top, the only constraints on the product will be the weight, as the desk may not be able to support too much load due to the hinges on the sides. The material of the desk shouldn't effect the usage of the product.



Data of the shelf

Length: 38 cm
Width: 34 cm
Height: 40 cm
Area available on shelf: 1292 cm squared
Volume available: 51680 cm cubed
Material: Plywood

The shelf seems to be able to fit any portable item used as a gambling tool. The other items on the shelf should not effect the product if the other items remain the same



Location

My product is going to be used **indoors** on one of my client's desk and be stored in a **shelf** when not in use

Ergonomics

In order for the product to be as **user friendly** and **efficient** as possible, it will have to be **small**, **light** and **portable** so that it can be easily moved from the clients' desk to the shelf with ease.

Even though there are many different places that the product can be **used** and **stored** in, the 2 most efficient places for storage and usage seem to be the desk and the shelf as they are **very close** to each other and will both seem to be able to support the product if it isn't too heavy or too big.



Evaluation

After collecting relevant data, I found out that the product has to be **smaller** than 34 x 38 cm and be shorter than 40 cm. I also found out that the product is most suited to be stored on a shelf and used on a desk. The product can't be too heavy as it has to be stored on shelf.

Research into Design Brief

Purpose of the interview

The purpose of the interview will be to receive feedback from both my clients for the current process of figuring out a suitable product. We will look at the past few pages for ideas, and questions will be asked for design specifications. The questions will be based on existing products, collection of data and inspiration board.

Existing products

Are there any products that fits your needs and wants?

“For me, all the products have a few good features that would fit our needs but none of them are perfect. The Saturn Roulette wheel seems like a great solution to our problem but is too large and expensive. On the other hand, the eDice seems to have a great price and has a good size but seems to be easily replaceable for a few real life dice and may not be very entertaining in the long term. In contrary to the other 2 products, the hardware random number generator looks terrible and wouldn't be very entertaining. As said in the slide, it seems to be more a decision maker than a solution to our problem.” -Sativik (older brother)

“In my opinion, the Saturn Roulette wheel is be the best solution as it seems very entertaining, but as Satvik said it's way too expensive. I don't think the other 2 products would be good solutions to our problem because the eDice just seems to be boring and the Hardware Random number generator seems to be very basic and easily replaceable.” -Varun (younger brother)

Are there any specific aesthetic considerations or design that you particularly prefer?

“I think that the roulette wheel and the eDice both look really nice. They both have an appealing modern design that I like. To me the material doesn't matter as long as the product is durable, but I would prefer a modern design.” -Sativik

“I agree with Satvik. I think a modern design would look cool.” -Varun

Inspiration board

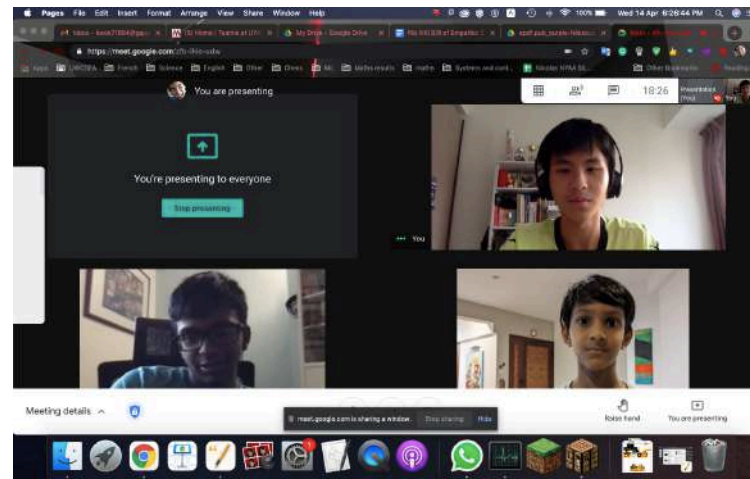
What do you like or dislike about the following solutions?

“Personally, my favourite solution is the mini roulette wheel at the bottom right of the page because it looks amusing, although a bit old. My least favourite would probably be the product at the top left because it takes up too much space, looks like it's designed for children and very fragile.” -Sativik

My favourite solution is the bingo machine above the roulette wheel because it looks unique. The slot machine also seems cool but may jam or break easily.” -Varun

What did I learn?

The interview helped me to recap my clients' needs and wants and allowed my clients to explain their preferences and dislikes about existing solutions. This allows me to produce a solution based on their preferences. In conclusion, I learnt that my clients prefer solutions that look more modern and solutions that have a large number of possible outcomes.



An interview with the clients

Inspiration board



Design Specification

Purpose

The purpose of the design specification is to **re-evaluate all the research** that has been conducted so far. The design specification consists of **specifications, justifications and method of testing**. The information will be split into 6 different factors: **function, aesthetics, materials, cost, size and safety**.

C2

Factors	Specifications	Justifications	Method of testing
1. Function	1.1 Solution must be a game that allows a form of betting between the clients 1.2 Product needs to have large number of possible outcomes to be entertaining	1.1 Clients needs a solution that solves their arguments about betting 1.2 Clients want the solution to be different to the old products and wants the solution to be entertaining	1.1 Ask for clients' opinions on whether the solution proposed is applicable or not 1.2 Ask for clients' opinion on whether or not they are satisfied with the number of possible outcomes and whether or not they think the product will be entertaining
2. Aesthetics	2.1 Product must have a modern feel 2.2 Product must be smooth and sleek 2.3 Product must not look dull and plain	2.1 Clients want the product to look appealing and is their preference 2.2 Clients want the product to look finished and well worked on 2.3 Clients want their product to not look boring and underwhelming	2.1 Do a visual test and ask for the clients' opinion 2.2 Do a visual and touch test and ask for the clients' opinion 2.3 Do a visual test and ask for the clients' opinion
3. Materials	3.1 Material must be durable and last over a long period of time 3.2 Material must have a modern style 3.3 Material must be hard	3.1 Clients specified that the material must be able to last long so the clients can reuse the product 3.2 Clients want the product to look appealing and is their preference 3.3 Material must not be able to bend or deform	3.1 Research about the material's durability 3.2 Do a visual test and ask for the clients' opinion 3.3 Do a touch test on the material and research whether it is applicable or not
4. Cost	4.1 Materials and circuit must be below 80 SGD	4.1 Clients wants a wide range of solutions but doesn't want to spend too much	4.1 Add a receipt for the clients to look at the total cost
5. Size	5.1 Product must be smaller than 34x38 cm 5.2 Product must be shorter than 40 cm 5.3 Product must not be too heavy . Lighter than 1 kg	5.1 Clients specified that the product must fit in the shelf above their desk 5.2 Clients specified that the product must fit in the shelf above their desk 5.3 Clients specified that the product must be portable and must not put too much of a load on the shelf	5.1 Measure the base of the product 5.2 Measure the height of the product 5.3 Measure the weight of the product
6. Safety	6.1 Product must not have sharp edges 6.2 Product must not be too heavy nor too light 6.3 Product should not have loose parts or loose circuits sticking out	6.1 Clients should not get hurt using the product 6.2 To prevent the product from falling off the shelf or table whilst in use and to reduce consequences after dropping 6.3 Client should not get hurt using the product	6.1 Touch test and visual test to check if there are any sharp corners 6.2 Measure the weight of the product. Judge to see the consequences of the product falling 6.3 Touch test and visual test to check if there are any loose parts or loose circuits

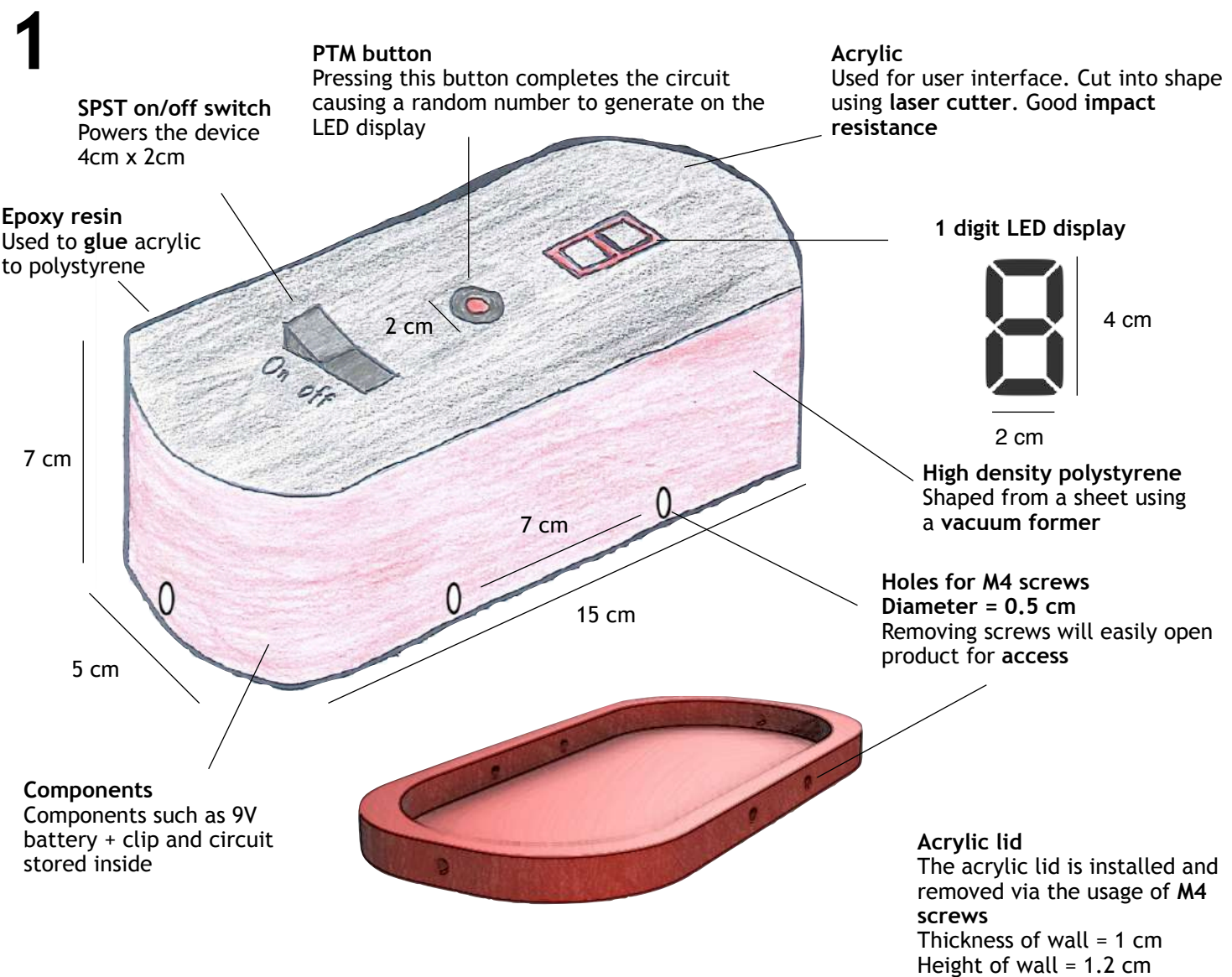
Exploration of Design Ideas

C3

Introduction

The purpose of the exploration of design ideas is to **evaluate possible design ideas** according to the **design specifications**. The green, yellow and red boxes are indicators on whether or not the idea has fulfilled the requirements regarding function, aesthetics, materials, size and safety. The evaluation gives a **justification** to the **evaluation** and gives a **general overview**.

■ Not met expectations ■ Met expectations ■ Exceeds expectations



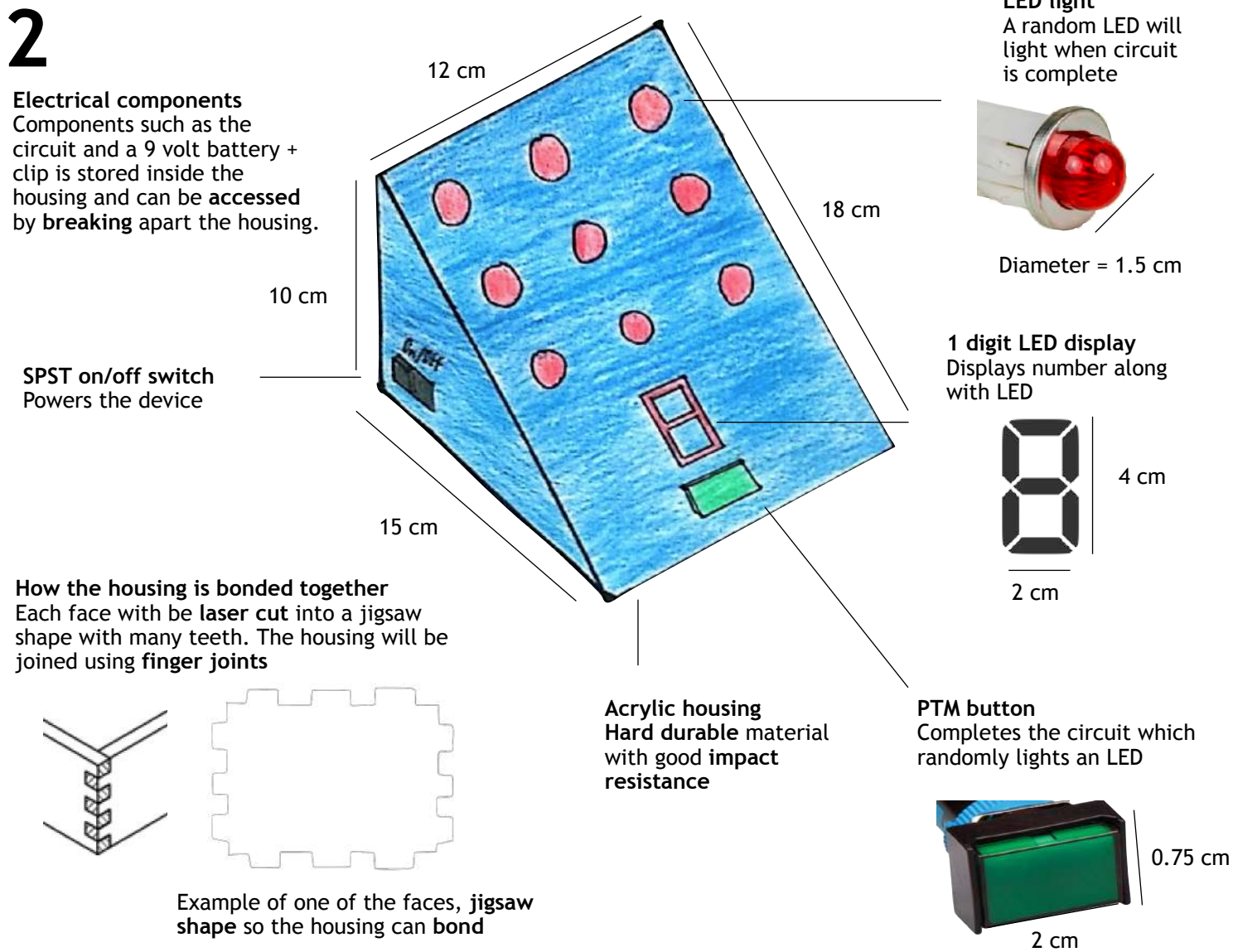
Evaluation

Explanation

- Function
- Aesthetics
- Materials
- Size
- Safety

What works well for this design is the **modern look** and the **smoothness** such as the **curved edges**. The size of the design is **compact** and should definitely fit inside the clients' shelf. The design should also **not** be too **heavy** and should be **portable**. In addition, there are **no sharp edges** and **lose parts** of the product. The electrical components are also easily **accessible** by the clients.

However, the product doesn't seem too **entertaining**. It seems more of a random number generator as opposed to a **betting tool**. In addition to this, the high density polystyrene used to make the housing **isn't as durable** as other materials in the **long term**.



Evaluation

Explanation

- Function
- Aesthetics
- Materials
- Size
- Safety

In this design, what seems to work well is the material choice. Acrylic is a **solid** thermoforming plastic that is very **durable**. The acrylic should also be **smooth** and shouldn't be rough. The size of the product is **reasonable** and the product shouldn't be too heavy.

On the other hand, the product doesn't seem to be **entertaining**. It appears to be more of a random number generator from 1-9, like the first product, with extra LEDs. There are also a few **sharp edges** on the corners which may cause some problems for the clients. One final weakness is the use of **finger joints**. Finger joints have to be **laser cut very precisely** or the design will be **weak**.

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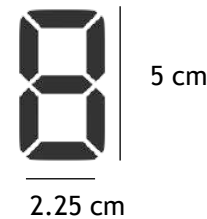
Wood screws

Wood screws are used to hold the user interface in place **protecting** the electrical components underneath



Plywood
Good strength to weight ratio,
good impact resistance

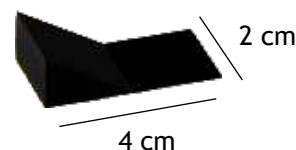
1 digit LED display
Displays a number for the game



PTM button
To reset, add or subtract from the LED display

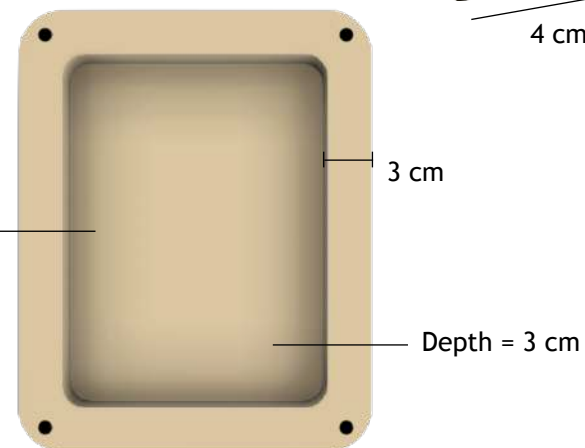


SPST on/off switch
Powers the device



Curved edges
Edges are sanded for **smoothness** and **safety** (no sharp edges)

Housing without user interface
Components such as the circuit and a 9 volt battery + clip will be **stored** here. These components will be **connected** to the user interface to make the product function.

**Evaluation Explanation**

- Function
- Aesthetics
- Materials
- Size
- Safety

What works well for this design is the large range of **possible outcomes**. There are also multiple ways to play this game and can also be used as a 2 digit counter. Plywood is also a very **durable** type of wood and has good impact resistance. Plywood also has a good **strength to weight ratio**. For the **safety** of the clients, most of the edges are also **smoothened** out to prevent any problems.

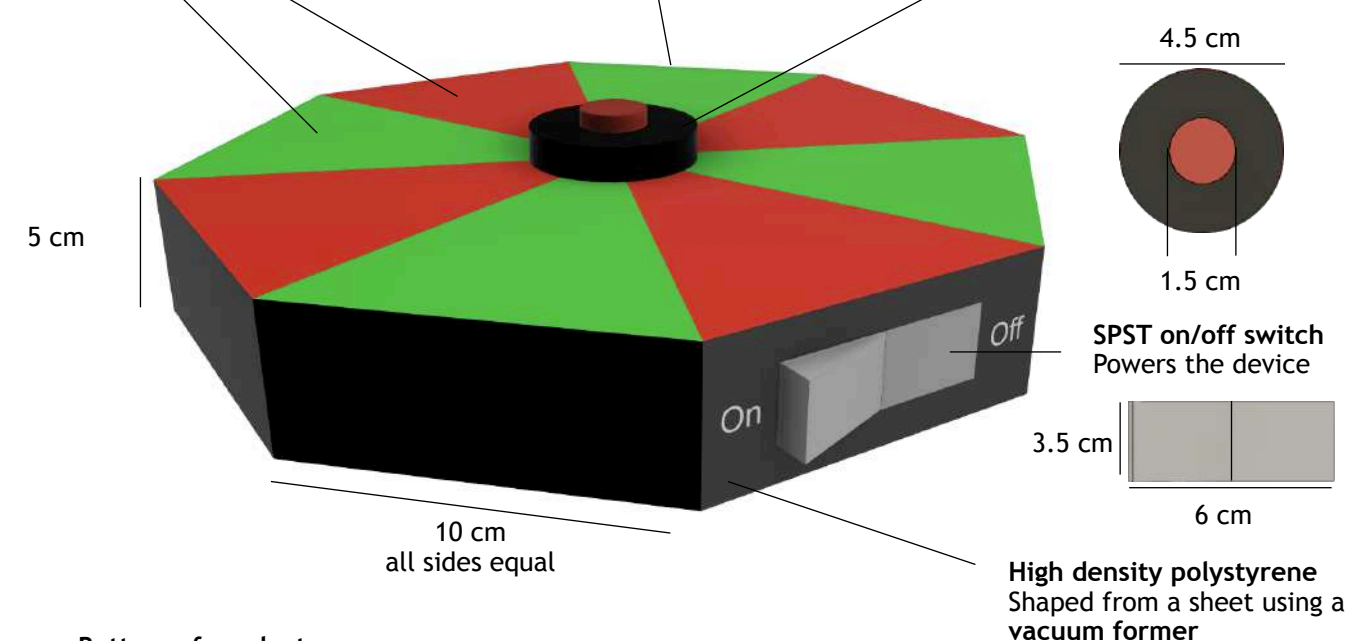
What could be improved might be the **colour** of the product. The housing only has **one colour** and may be too **plain** for the clients' liking. The product also **isn't as compact** as the other products and may be deemed as too **large** even having fit in the clients' shelf.

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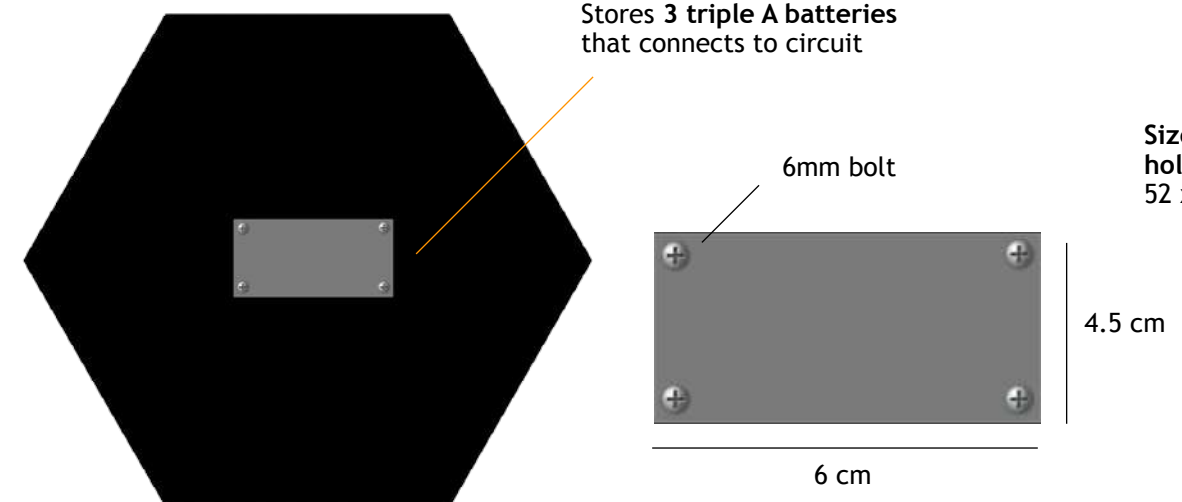
LED light panel
One panel **randomly** lights up when button is released after cycling

Epoxy resin
Used to **glue** LED panels to the high density polystyrene housing

PTM button
Completes the circuit which causes LED panels to cycle and stops on a **random** panel when button is released



Bottom of product



Size of standard battery holder for 3 triple A cells
52 x 37 x 13 mm

Evaluation Explanation

- Function
- Aesthetics
- Materials
- Size
- Safety

The main thing that works well in this design is the **modern** look and the overall completed **sleek** aesthetic of the design. Although there are only 6 possible outcomes in this design, the **cycling** and the **roulette design** may make up for the small amount of outcomes and may be moderately **entertaining**. The location of the triple A batteries is also very **efficient** for the client in the need of **changing** the batteries.

However, on the contrary, the high density polystyrene used in this design isn't **durable** and might be **flimsy**. The **size** and **weight** of the product **isn't too large** although the design could be **more compact**. One major issue about this design is the **location** of the on/off switch. This loose part is sticking out and can be powered on and off easily by **accident**.

C3

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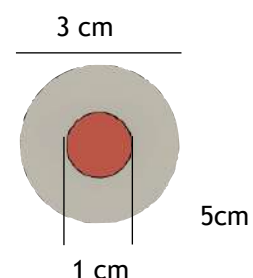
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Triple A batteries

Triple A batteries are used to power the device and is stored underneath

PTM button

One button to add a **random** amount one to subtract a **random** amount

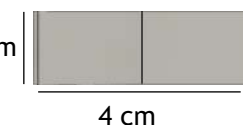


LED



5 mm diameter LED

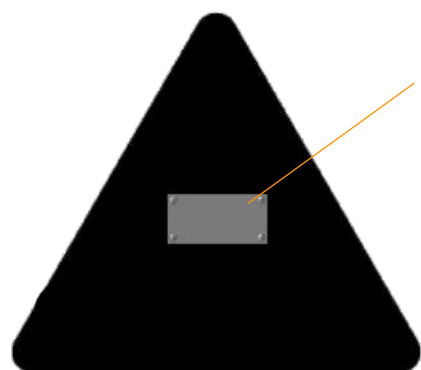
SPST on/off switch Powers the device



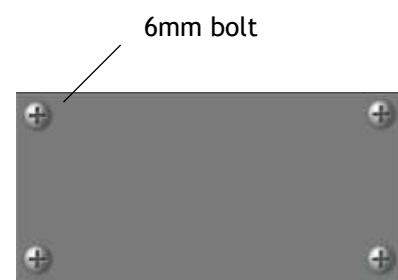
Acrylonitrile Butadiene Styrene (ABS) Strong and hard, can be heated and moulded easily

20 cm All sides equal

Bottom of product



Stores 3 triple A batteries that connects to circuit



6 cm

Size of standard battery holder for 3 triple A cells 52 x 37 x 13 mm

Evaluation

Explanation

- Function
- Aesthetics
- Materials
- Size
- Safety

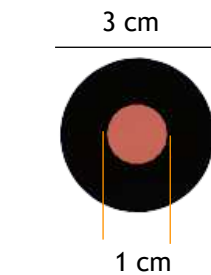
In this design, what works out well is the way the product **works**. There should be **different** ways to **play** this game and it's **unique** which should make the device **entertaining** and **new** for the clients. The product also has a **modern** look that appears **finished** and **smooth** and material used is also **hard** and **durable**.

What could be improved is making the product a bit more **compact**. Another disadvantage to this design is the **weight** of the product. Due to its large **size** and a large amount of ABS used for its **rigidity**, the **weight** of the product may be a bit **higher than desired**.

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PTM button

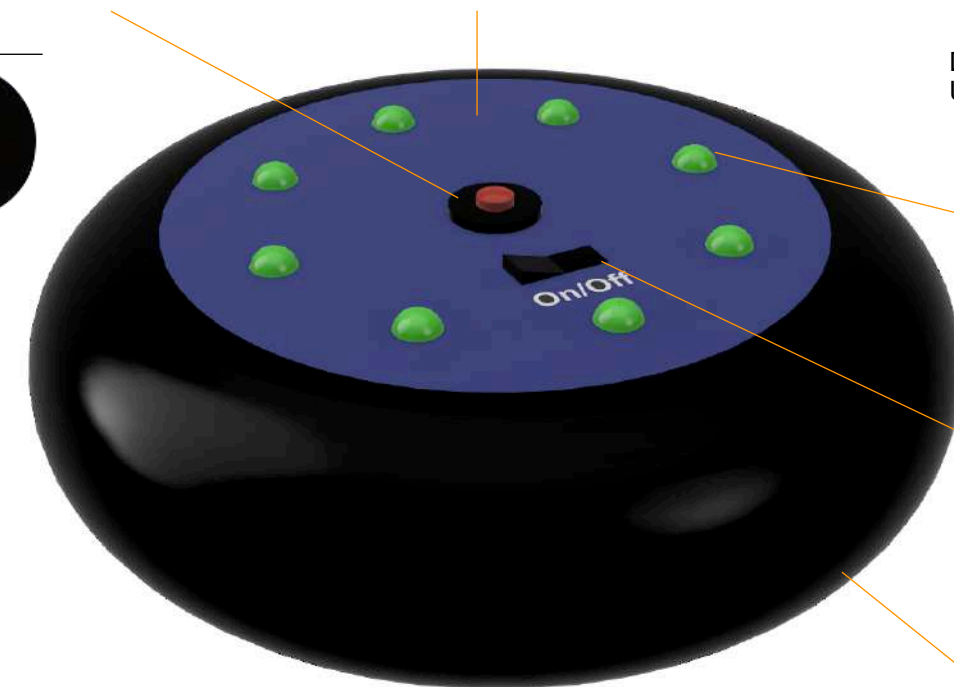
Completes the circuit which causes LEDs to **cycle** and stops when the button is released



6 cm Height

Acrylic user interface glued to housing

User interface can be cut using a **laser cutter** and **glued** to housing from the inside using ethylene-vinyl acetate (EVA) from a **hot glue gun**. The diameter of user interface will be slightly **larger** than the diameter of the gap in the housing since the user interface is being **glued** from inside the housing.



Diameter of UI = 11 cm

Diameter of gap = 9 cm

LED



5 mm diameter LED

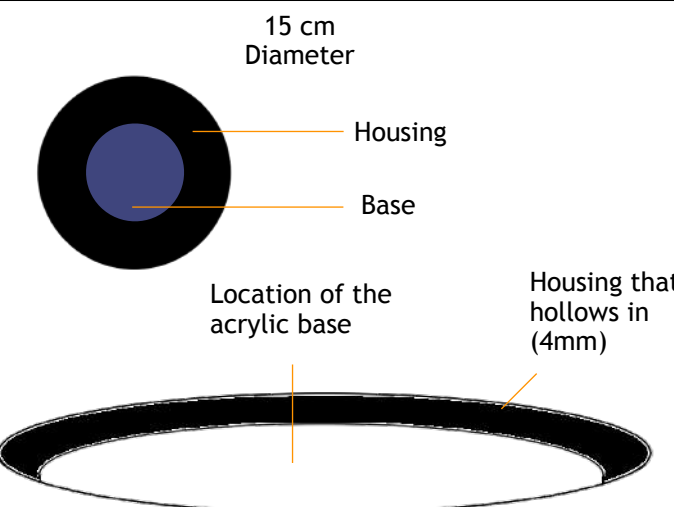
SPST on/off switch Powers the device



High density polyethylene Durable and good impact resistance

Bottom of product

The bottom of the product will have a **similar** design to the top, however the size of the gap will have to be **larger** in order to allow the user interface to fit through when making the product. The acrylic will also be stuck to the bottom from the outside. For the base to fit nicely, the housing at the bottom will slightly **hollow** inwards.



15 cm Diameter

Housing

Base

Location of the acrylic base

Housing that hollows in (4mm)

Diameter of base = 13 cm

Diameter of gap = 11 cm

Evaluation

Explanation

- Function
- Aesthetics
- Materials
- Size
- Safety

What works well for this design is the **modern aesthetics** and the **completeness** in this design. Polyethylene is an excellent choice as it is a very **durable** material and has **good impact resistance**. The product is also quite **lightweight** and there are **no sharp edges**.

What could be changed is the **number of possible outcomes**. The design can be made more **entertaining** by having a **larger range of outcomes**. However, increasing the number of possible outcomes will lead to an **increase in LEDs** which may lead to an **increase in size of the product**.

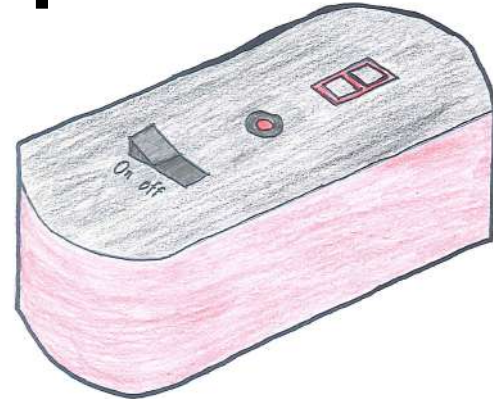
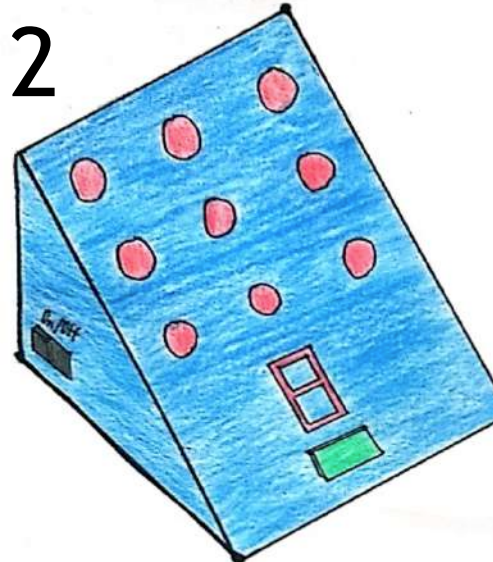
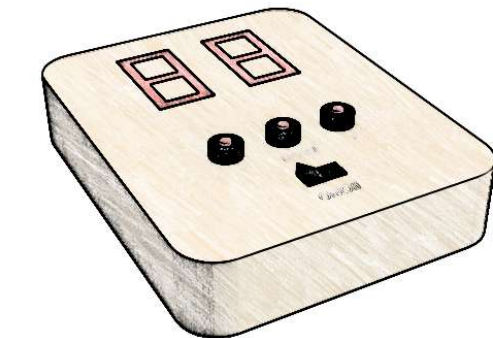
Client and Self Review of Design Ideas

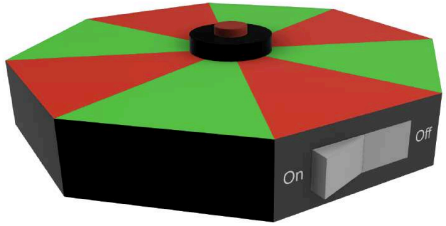
The following 2 pages will discuss the 6 design ideas above and compare them to the design specifications. Each design specification will be rated from 0-2 (0; not met specification, 1; met specification, 2; exceeds specification). The total will then be added up for the whole design idea for a score. A self and client review will be concluded and written down. Lastly one design will be chosen and highlighted. This design will then move on to the next stage of the project.

0 = not met expectations

1 = met expectations

2 = exceeds expectations

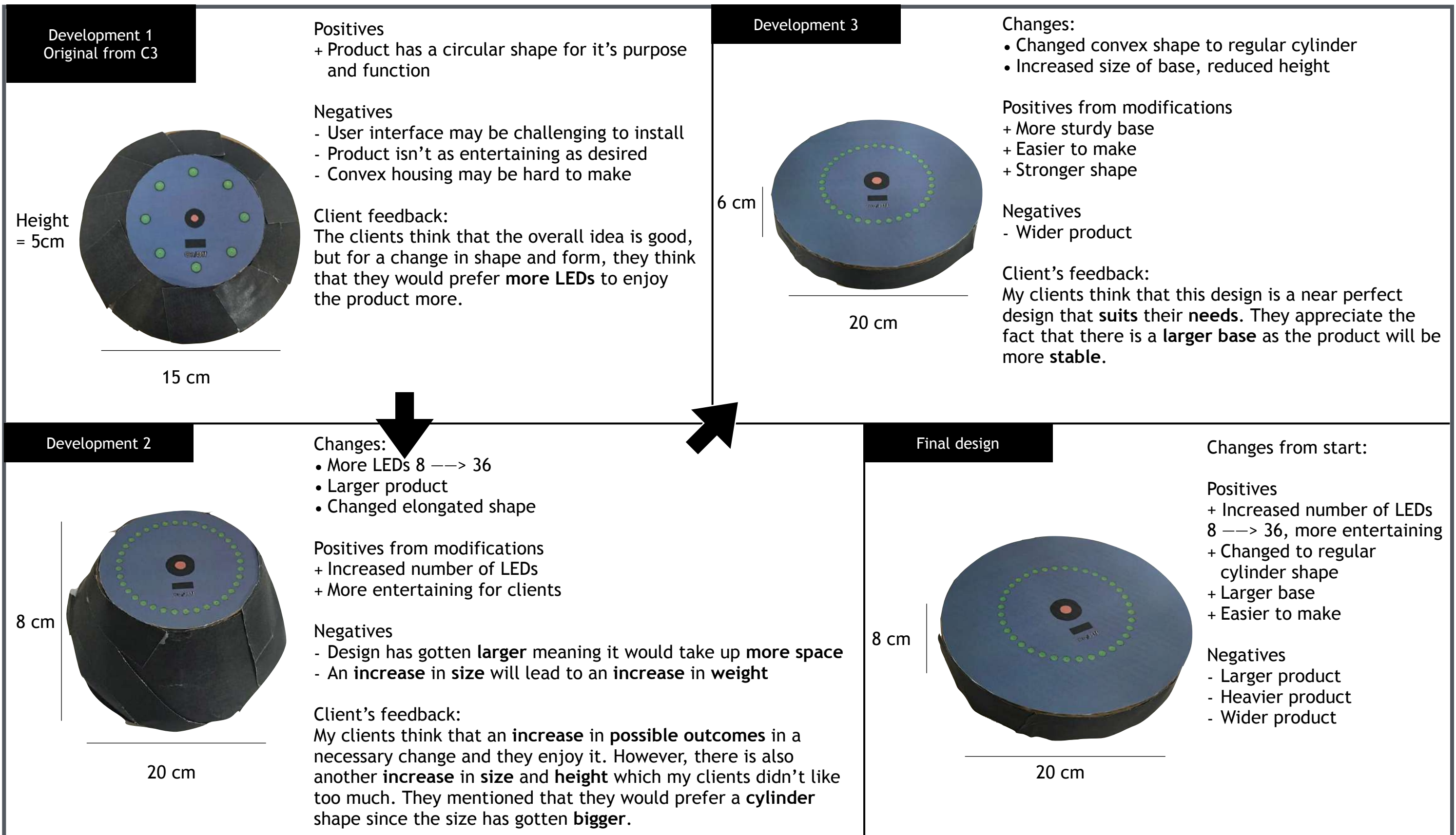
Design	Specification points			Score	Self evaluation + Client evaluation
1 	Function 2 Solution must be a game that allows a form of betting between the clients 0 Product needs to have large number of possible outcomes to be entertaining	Materials 0 Material must be durable and last over a long period of time 1 Material must have a modern style 0 Material must be hard	Safety 2 Product must not have sharp edges 1 Product must not be too heavy nor too light 2 Product should not have loose parts or loose circuits sticking out	18/28	Self evaluation: In my opinion, this design has a cool unique shape that seems appealing and compact , however, for the product to improve, the material of the housing needs to be more durable . I feel as though this solution won't suit my clients. Client evaluation: The clients say that the design seems to have a good look however they both agreed that the product doesn't suit their needs and would be quite boring to use.
2 	Function 2 Solution must be a game that allows a form of betting between the clients 0 Product needs to have large number of possible outcomes to be entertaining	Materials 2 Material must be durable and last over a long period of time 1 Material must have a modern style 2 Material must be hard	Safety 0 Product must not have sharp edges 2 Product must not be too heavy nor too light 1 Product should not have loose parts or loose circuits sticking out	18/28	Self evaluation: This design seems compact and durable . There is also a unique shape compared to other products, on the other hand, finger joints may be quite weak and I don't think the product would suit my clients. Client evaluation: The clients like the colour of the product however, like the first product, there are too little outcomes and this product won't suit their needs.
3 	Function 1 Solution must be a game that allows a form of betting between the clients 2 Product needs to have large number of possible outcomes to be entertaining	Materials 2 Material must be durable and last over a long period of time 0 Material must have a modern style 2 Material must be hard	Safety 2 Product must not have sharp edges 2 Product must not be too heavy nor too light 1 Product should not have loose parts or loose circuits sticking out	18/28	Self evaluation: I feel like this design would be quite enjoyable for the clients because the product seems more like a game as opposed to a betting tool. Client evaluation: One of the clients think that this design suits their needs and thinks it's a really cool whereas the other thinks the product is mediocre and doesn't like the bland colour of the wood.

Design	Specification points			Score	Self evaluation + Client evaluation
4 	Function 2 Solution must be a game that allows a form of betting between the clients 0 Product needs to have large number of possible outcomes to be entertaining	Materials 0 Material must be durable and last over a long period of time 2 Material must have a modern style 0 Material must be hard	Safety 1 Product must not have sharp edges 1 Product must not be too heavy nor too light 0 Product should not have loose parts or loose circuits sticking out	15/28	Self evaluation: I feel like this design has a cool concept but seems to feel very limited in terms of the versatility of usage due to it's small range of outcomes . Client evaluation: The clients agree with me that this is a cool concept, however there are a lot of faults with this design such as the on/off switch sticking out and the durability of the materials .
	Aesthetics 2 Product must have a modern feel 1 Product must be smooth and sleek 2 Product must not look dull and plain	Size 1 Product must be smaller than 34x38 cm 2 Product must be shorter than 40 cm 1 Product must not be too heavy. Lighter than 1 kg			
5 	Function 1 Solution must be a game that allows a form of betting between the clients 2 Product needs to have large number of possible outcomes to be entertaining	Materials 2 Material must be durable and last over a long period of time 2 Material must have a modern style 2 Material must be hard	Safety 2 Product must not have sharp edges 1 Product must not be too heavy nor too light 2 Product should not have loose parts or loose circuits sticking out	22/28	Self evaluation: Overall, I really enjoy the aesthetics of this product. Like the 3rd product, I feel like the clients would enjoy using this product since it has more of a gaming feel to it. Client evaluation: Both the clients think that this would be a great solution to their problem because of the way the product works . They also stated that they find the aesthetics appealing .
	Aesthetics 2 Product must have a modern feel 2 Product must be smooth and sleek 1 Product must not look dull and plain	Size 1 Product must be smaller than 34x38 cm 1 Product must be shorter than 40 cm 1 Product must not be too heavy. Lighter than 1 kg			
6 	Function 2 Solution must be a game that allows a form of betting between the clients 1 Product needs to have large number of possible outcomes to be entertaining	Materials 2 Material must be durable and last over a long period of time 2 Material must have a modern style 2 Material must be hard	Safety 2 Product must not have sharp edges 1 Product must not be too heavy nor too light 1 Product should not have loose parts or loose circuits sticking out	25/28	Self evaluation: In my opinion almost everything works out well in this design and the product definitely suits the clients' needs. The only thing that needs improvement is the number of possible outcomes that needs to be increased. Client evaluation: Both my clients think that this is an amazing solution to their problem. Agreeing with what I said above, they also think there needs to be more LEDs in the product.
	Aesthetics 2 Product must have a modern feel 2 Product must be smooth and sleek 2 Product must not look dull and plain	Size 2 Product must be smaller than 34x38 cm 2 Product must be shorter than 40 cm 2 Product must not be too heavy. Lighter than 1 kg			

Development of Shape and Form

C4

After having selected a design from the page before, there were certain **modifications** that were noted by both the clients and I. This led to the creation of multiple **prototypes** using cardboard aiming to solve the problems. My thoughts and **feedback** received from the client will be what changes each prototype.



Development of Materials and Finishes

C4

This page will compare **multiple** options for **materials** and look into the **advantages** and **disadvantages** for each option. I will then **select** an option, **highlight** it in a green box, and explain my **reasoning** behind each choice.

Body

Choice of materials

Wood



- Low thermal conductivity
- Electrical insulator
- Good aesthetics
- Good strength to weight ratio
- Wood will swell in humid environments
- Some woods isn't water resistant
- Can be vulnerable to insects and fungi

Acrylic



- Many colours available
- Lightweight
- Easy to engrave
- Resistance to weathering
- Good strength to weight ratio
- Can be scratched easily
- Easily stained
- Subject to cracking

3D print (PLA)



- Derived from a renewable resource
- Compostable in special facilities
- Easily 3D printed, easy to make
- Variety of different shapes and figures can be printed
- High precision



Choice of wood coating

Beeswax



- Easy to apply
- Can be used with any type of wood
- Little protection against scratches and weathering

Teak oil



- Water resistant
- Aesthetically pleasing
- East to apply
- Scratches easily
- Little protection to wood

Osmo oil



- Resistant to dirt
- Smoothens surface effectively
- Provides protection to surface
- Might have to be reapplied after a period of time

Reasoning behind choice

I have decided to choose kapok with osmo oil because kapok is a **strong durable** wood that looks **aesthetically pleasing**. In addition to this, osmo oil **works well alongside** with Kapok, is **easy to apply** and provides good protection.



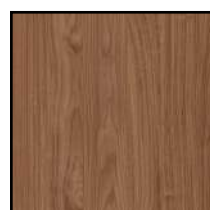
Choice of woods

Teak



- Strong
- Very durable
- Dark material
- Expensive
- Difficult to glue together

Kapok



- Super hard and strong
- Good strength to weight ratio
- Moisture resistant
- Resilient and withstands shocks

Pine



- Cheap
- Lightweight
- Resists shrinking and swelling
- Can be easily damaged
- Vulnerable to scratches and dents

User interface

Choice of materials

Wood



Acrylic

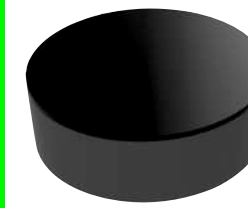


3D print (PLA)

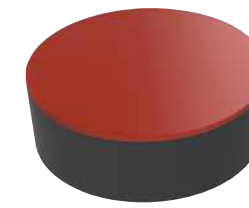


Choice of colour

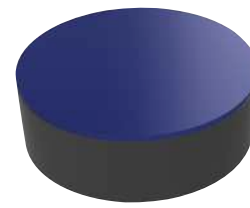
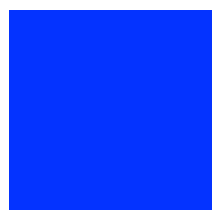
Black



Red



Blue



Reasoning behind choice

I have decided to choose **black** for my user interface because of the **aesthetics** and how I feel it will **match** well with the wood. A black user interface will also provide a good **contrast** with the circuit components resulting in the circuit components **standing out more**.

Final design + conclusion

The final design features **kapok** with **osmo oil** as the housing and a **black acrylic sheet** as the user interface. I feel as though this will create a nice **contrast of colours** and will look **aesthetically pleasing** for the final product.



Development of Joining and Construction Methods

C4

This page will explore the different types joining and production methods for joining my user interface to my housing. The green box indicates the chosen joining/production method to be taken forward.

Joining method (user interface)

Option 1 wood screws:

Screws can be used to hold the user interface in place. They can be easily added and removed for accessing electrical components in the housing. There are also many different types of screws for different applications, however the screws I will use will have a.

Pros

- + Can be reused
- + Can access electrical components easily
- + Good strength
- + Many different shapes and sizes

Cons

- Can rust
- Can harm material if not inserted properly
- May be deemed unattractive
- Takes precision to make

Wood screws



Option 2 resin:

Resin is another option to hold the user interface in place. It is quick to apply and will form a strong adhesive bond between the user interface and the housing. If used correctly, resin will not damage the aesthetics.

Pros

- + Cannot be seen
- + Strong
- + Easy and quick to apply
- + Low cost

Cons

- Single use
- Might have to be reapplied after a period of time
- Needs to be precisely applied with no excess amount



Resin will be applied along this face



Type of joint user interface to housing

Option 1 butt joint:

A butt joint is a technique in which two pieces of material are joined by simply placing their ends together without any special shaping. Simple but weak.



Option 2 lap joint:

A lap joint or overlap joint is a joint in which the members overlap. Lap joints are stronger than the simple butt joints but requires a bit of shaping.



Option 3 finger joint:

A finger joint is a type of joint formed by interlocking cut out parts from 2 interlocking pieces of material. This joint is strong but requires a lot of precision.



Reasons justifications and final design

Joining method: screws

I chose to use screws as my joining method because although it may be unattractive, using screws provides the clients the opportunity to access the electrical components in the housing. On top of that, the strength provided by the screws is enough to keep the user interface stuck to the housing.

Type of joint: Butt joint

I chose to use butt joints as my type of joint because it is simple to make and the least time consuming.

How the housing is going to be joined and made

Option 1 base + housing

Base + one complete layer of the main body. This method may look better however it will take a long time to make.



Option 2 base + multiple layers of housing

4 equal layers including base (20mm each layer) of housing. This method will look worse however, I will use a CNC milling machine to save time.



Housing type

I chose to use multiple layers since it will save time and is easier to make. The 4 layers and the base will be glued together to form the housing.

Wood screws
Butt joint



3 layers + base

Development of Circuit

C4

This following page will look into different types of circuits. Green boxes will indicate the chosen option

Option 1: Electronic roulette kit Model AK-300

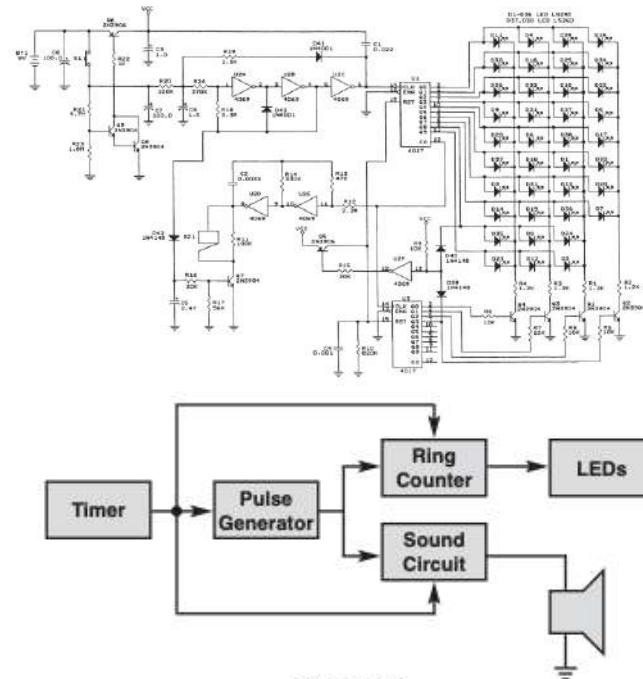


Figure 1

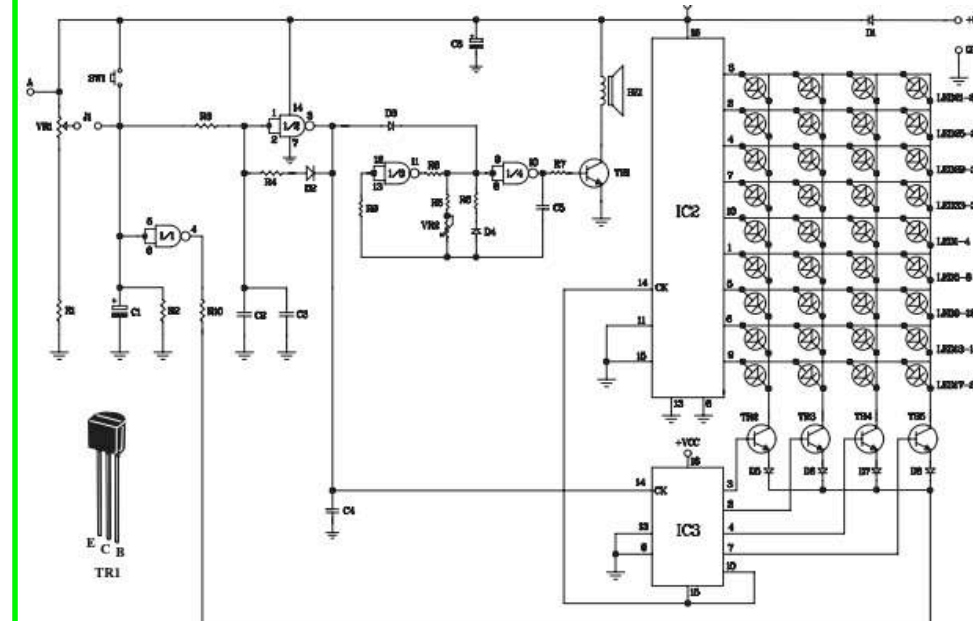
The circuit

This is a circuit representing a 36 LED roulette wheel. Figure 1 represents a simplified version of the main circuit diagram. The Timer is used to power all the circuits. The Pulse Generator creates pulses that generate a sound and forces the ring counter to move the position of the lit LED. The Sound Circuit generates the sound of a bouncing ivory ball for realism, and produces a warning sound before the device is powered off. Lastly, the Ring Counter lights each LED in a circular sequence. The LEDs represent the position of the ivory ball. In addition, there is an automatic power off system when the circuit has not been used for a certain period of time

Pros and cons

- | | |
|---|---------------------------|
| + Circuit has an automatic power off system | - Volume cannot change |
| + Circuit produces sound | - Many circuit components |
| + Circuit cycles between LEDs | - Fairly large circuit |
| + Good range of possible outcomes | - A few excess features |

Option 2: Future Kit 126 SUPER ELECTRONIC ROULETTE 36 DOT WITH SOUND



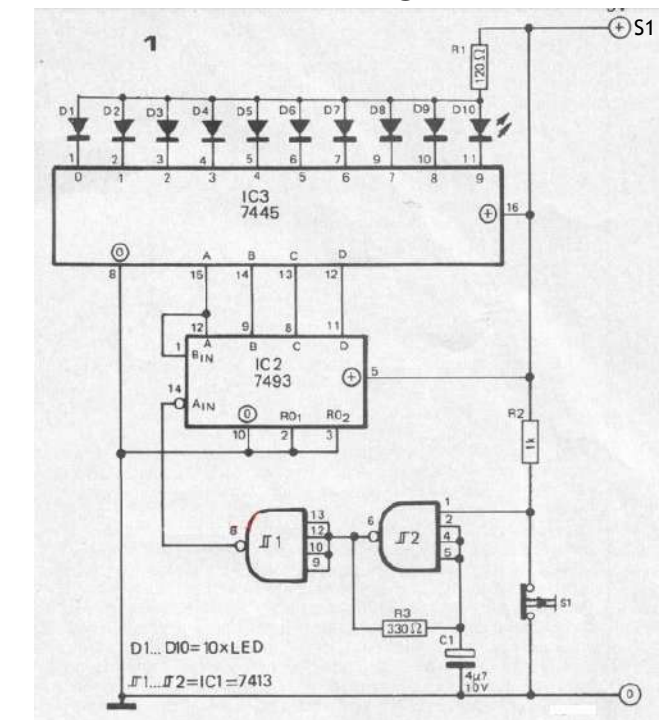
The circuit

This circuit is similar to the first however, with a few less features. When a button is pressed, IC1/2 generates a frequency which is by pin CK of IC3. IC3 will then start counting from 3, 2, 4, 7 sequentially. Every time pin 10 is reached, IC2 will start counting throughout the rows of LEDs and IC3 will reset itself and begin counting from 3 again. IC2 resets if all 36 LEDs have been powered once. This is repeated until the button is released C1 will discharge a voltage which when received by IC1/2 will slowly generate a lower frequency until only 1 LED is remained lit. A buzzer will also sound when the circuit is running. The client can change the cycling rate using the potentiometer.

Pros and cons

- | | |
|-----------------------------------|------------------------|
| + Easy to use and understand | - Fairly large circuit |
| + Cycling rate can be changed | - Volume cannot change |
| + Good range of possible outcomes | |
| + Circuit produces sound | |

Option 3: Wheel of Fortune game circuit



The circuit

The LEDs on this roulette wheel starts cycling when the switch, S1 is pressed. This results in pin 1 of IC 7413 going high. Then, there is an oscillator at IC 7413 which is started and wave pulses begin switching pin 14. This 14th pin is part of a four bit counter IC2. When S1 is released the oscillation comes to a halt. IC2 will also count the number of pulses/cycles of LEDs the system goes through while the switch, S1 is pressed. This will detect whether or not a player is cheating by waiting for the cycle and holding the switch for too long.

Pros and cons

- | | |
|-----------------------------|--|
| + Small and compact circuit | - Low range of possible outcomes |
| + Few circuit components | - No cycling after S1 is released resulting in possible exploits |
| + Simple construction | - No sound |
| + Super easy to understand | |
| + Anti cheat system | |

Summary

In conclusion I decided to pick option 2 because the **positives** simply **outweigh** the negatives. The circuit fits my clients' needs more than the other 2 options and is **simple** and **easy** to understand. Circuit 1 is a more **sophisticated** version of circuit 2 however, I feel as it is too **complex** and with many **additional features** that are **not needed** such as the automatic power off system or the warning buzzer before power off. Circuit 3 on the other hand, is a design that is simple and easiest to understand however, there are a few downsides such as a **smaller range of possible outcomes** and most importantly the immediate **halt of cycling** after the S1 switch is released, leading to possible **exploits** from the clients. Although there is an anti cheat system countering this mechanism I believe that it is pointless as it is just covering up a missed feature that circuit 1 and 2 both have.

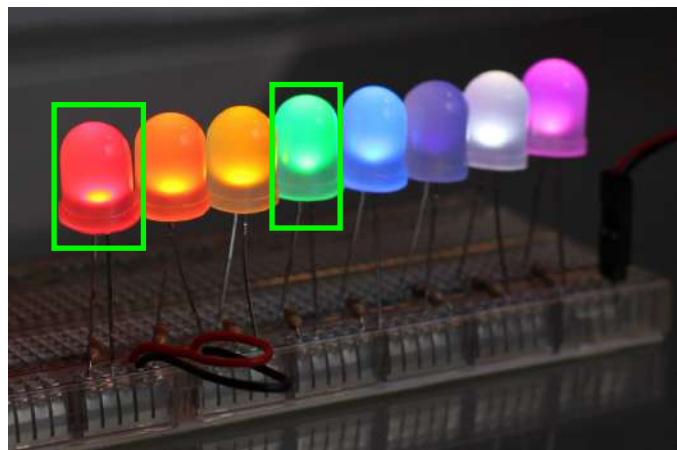
Development of Circuit Components and User Interface

C4

This following page will look into different types of circuit components and user interface layout. Green boxes will indicate the chosen option

Circuit components

Colour of LEDs



For my choice of LEDs I have decided to choose red and green to replicate a roulette theme and am planning to arrange the LEDs in an alternating pattern.



Rocker switch

Choice of SPST switch



Latch switch

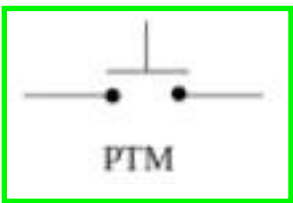


Toggle switch



Slide switch

Choice of button



PTM

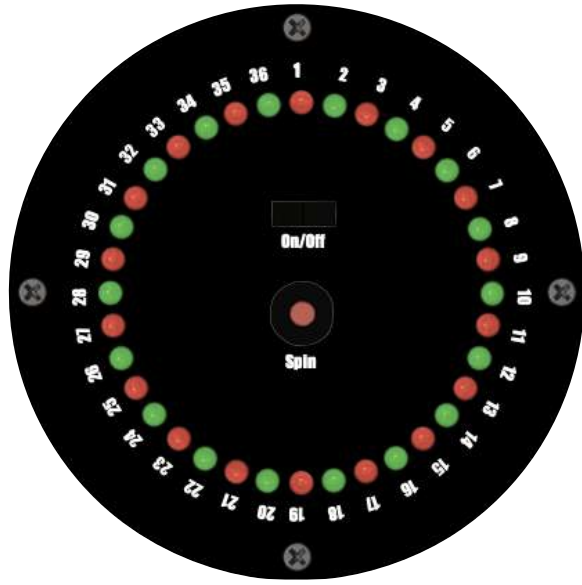


PTB

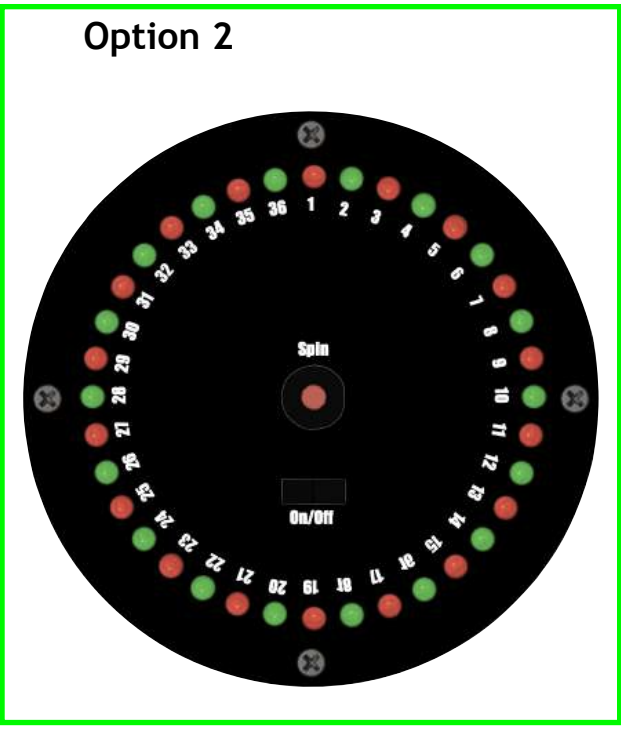
For my circuit to work, a PTM switch is required to complete the circuit. A red PTM button will fit well with the colour scheme of the product.

User interface

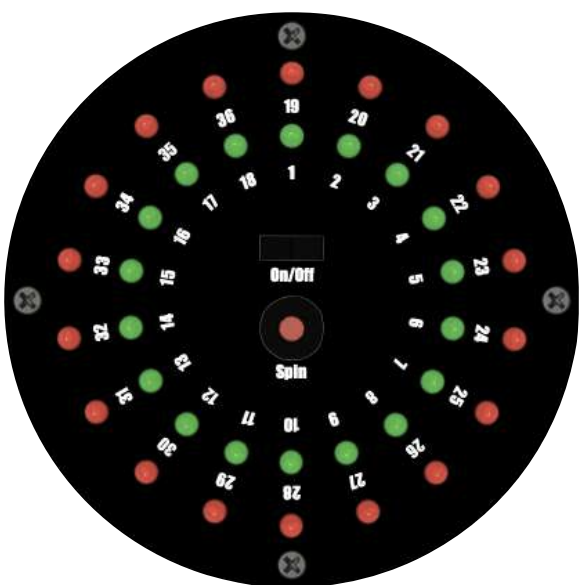
Option 1



Option 2



Option 3



Reasoning behind choice

I have decided to choose option 2 because I feel it looks the best and the most “correct”, such as the PTM button in the middle with the on/off switch below it. In my opinion, having the numbers on the inside of the ring of LEDs looks more aesthetically pleasing compared to having the numbers on the outside, which is another reason why I chose option 2 over option 1.

Final design

8 cm
(4 x 2cm)



20 cm

Overall summary

Throughout the past few pages, I have developed my product in many ways. I am very happy with my results as I feel like the developed version looks a lot more complete, finished and well thought out compared to the initial design.

Planning for Production

For the following page, I will be going over steps required in order to make the product, the equipment needed, estimated time, quality control measures and safety

C5

Stage number	Part/component + name of process	Equipment needed	Description of process	Estimated time	Quality control measure/health safety
1	Circuit Installing circuit components	- Circuit board - Circuit components (resistors, transistors, capacitors, e.t.c.)	Insert components in corresponding place on circuit board, follow instructions on manual	10 mins	- Carefully follow instructions - Ignore components that are going to be connected with a wire
2	Stripping wires	- Wires - Wire cutter	Strip required amount of wires needed for external components such as buttons or switches with a wire cutter	10 mins	- Don't strip too much of the wire, more can always be stripped later
3	Soldering circuit components	- Circuit board and components - Male + female wires - Wire cutter - Solder - Soldering iron - Soldering station	Solder the circuit components to the circuit board using soldering equipment . Use the wire cutter to cut off the excess wire holding the circuit component in place	40 mins	- Avoid burning yourself - Avoid inhaling smoke released from solder - Wear goggles - Consider using the vice to clamp the edge of the circuit board for more efficiency and precision - Copper on circuit shouldn't be visible after soldering - Ensure each component is in correct place. Refer to manual if needed
4	Soldering external components to wires	- External circuit components - Wires - Solder - Soldering iron - Soldering station	Solder external components such as switches and buttons to wires using soldering equipment	5 mins	- Avoid burning yourself - Avoid inhaling smoke released from solder - Wear goggles - Consider using the vice to clamp components for more efficiency and precision - There should be a sufficient amount of solder for a firm joint
5	Installing LEDs	- Circuit board with male + female wires - Red and green LEDs - Hot glue gun	Attach LEDs to male female wires , put the longer side of the LED into the positive terminal and the shorter side into the negative terminal in an alternating colour order . Hot glue LED to wire	5 mins	- Ensure the LED is installed correctly, positive and negative side - Ensure each LEDs are installed in an alternating order - Ensure LEDs are fastened correctly with hot glue

Planning for Production

C5

Stage number	Part/component + name of process	Equipment needed	Description of process	Estimated time	Quality control measure/health safety
6	Soldering external components to circuit board	- Circuit board - External components with wires attached - Solder - Soldering iron - Soldering station	Solder each external circuit component with wires in corresponding location to circuit board using soldering equipment	10 mins	- Avoid burning yourself - Avoid inhaling smoke released from solder - Wear goggles - Consider using the vice to clamp the edge of the circuit board for more efficiency and precision - Copper on circuit shouldn't be visible after soldering - Ensure each component is in correct place. Refer to manual if needed
7	Test circuit	- Circuit board with soldered components - Power source connected to circuit	Power the circuit and test , if the circuit works then keep going, if not, troubleshoot and solve the issue	2 mins	- Use the correct power source, correct voltage
8	Housing Creating file for CNC milling machine	- Computer - Fusion 360 software	Use Fusion 360 to create a sketch of profile of multiple housing layers and export as a DXF file for the CNC milling machine	10 mins	- Beware of units - Ensure the sketch is the correct size, 1:1 to the real product
9	Measuring dimensions of kapok plank	- Kapok wood plank - Ruler/measuring tape	Check the dimensions of kapok plank, make sure there is enough space for the housing to be cut out. Ensure the height is 200mm as the height isn't being taken into account by the CNC milling machine	5 mins	- Use correct ruler and check correct units

Planning for Production

C5

Stage number	Part/component + name of process	Equipment needed	Description of process	Estimated time	Quality control measure/health safety
10	Cutting the housing	- CNC milling machine - Plank of kapok wood - DXF file	Send the DXF file to the CNC milling machine , then place the kapok plank in the correction position before starting up the machine. Repeat until all parts of the body are done	15 mins	- Plank must be in correct position - Stay near machine and be ready to press the stop button in case anything goes wrong - Get supervision - Don't stand too close to the machine - Be aware of loose parts
11	Assembling the housing	- All parts of housing - Resin - Brush - Clamp	Apply resin evenly on the intersections of the housing with a brush . Use a clamp to hold the housing in place and wait for the resin to take into effect	10 mins	- Ensure the resin is evenly spread - Clamp the housing tightly in the correct position
12	Sanding the housing	- Kapok wood housing - Disk sander - Sand paper	Sand down the edges of the housing with the disk sander	10 mins	- Sand evenly - Be careful with fingers
13	Applying finishing to the housing	- Kapok wood housing - Osmo oil - Polishing cloth - Drying rack	Apply osmo oil onto surface of wood using polishing cloth , then leave on drying rack to dry	10 mins	- Osmo oil should be spread even - Avoid getting the osmo oil off the wood
14	Marking out	- Top layer of kapok housing - Pencil - Ruler - Masking tape	Use a ruler to measure the desired positioning and dimensions of the hole on the top layer of the kapok wood housing. Place masking tape on the area and use a pencil to mark the spot for future drilling	5 mins	- Be very precise - Be sure to use the masking tape to protect the wood
15	Drilling the hole	- Marked out top layer of housing - Drill with correct diameter - Clamp	Use a clamp to hold the housing in place, then, drill hole using drill on marked out spot to a desired depth	5 mins	- Be very precise, there's only one shot - Ensure the housing is clamped tightly - Use correct diameter drill
16	User interface Creating the file for the user interface for 3D printing	- Computer - Fusion 360 software or adobe illustrator	Use Fusion 360 or Adobe Illustrator to create a sketch of the user interface and export as a file for the laser cutter	20 mins	- Beware of units - Ensure the sketch is the correct size, 1:1 to the real product - Remember to include small details

Planning for Production

C5

Stage number	Part/component + name of process	Equipment needed	Description of process	Estimated time	Quality control measure/health safety
17	Creating the user interface	- Laser cutter - Black acrylic - User interface file	Position the acrylic in place, then send the DXF file to the laser cutter and begin laser cutting . Remove acrylic when done	10 mins	- Position acrylic correctly - Don't open the machine before process is done
18	Polishing up the user interface	- White marker	The cut out user interface should've come with a protective layer and the laser cutter should've cut out all the details such as text and holes. Use a white marker to colour over all text on the protective layer	10 mins	- Use the correct coloured marker - Don't peel protective layer too early
20	Test to see if the user interface fits	- Wood screws - Screwdriver	Insert and remove wood screws to test if the user interface and housing align, if it doesn't redo user interface	2 mins	- Use correct sized screw and screwdriver
21	Finishing Installing electrical components into user interface	- User interface - PTM switch - Rocker switch	Insert and fix electrical components into the user interface	2 mins	- Fix component in correct location
22	Soldering circuit to circuit components	- User interface with circuit components - Circuit - Solder - Soldering iron - Soldering station	Solder circuit components to circuit using soldering equipment	5 mins	- Avoid burning yourself - Avoid inhaling smoke released from solder - Wear goggles - Consider using the vice to clamp the edge of the circuit board for more efficiency and precision - Ensure each component is in correctly wired to circuit. Refer to manual if needed
23	Placing LEDs in user interface	- Completed circuit + user interface	Push LEDs through holes of user interface corresponding to the number	3 mins	- Be precise - Avoid breaking the connection between LEDs and male female wires
24	Assembling the product and final test	- Completed circuit + user interface + battery - Housing - Wood screws - Screwdriver	Place circuit inside the housing and user interface on top of the housing. Screw the user interface to the housing using wood screws and a screwdriver , then test the product	2 mins	- Use correct sized screw and screwdriver

Presentation Drawing

The presentation drawing presents a realised rendered product made digitally on Fusion 360

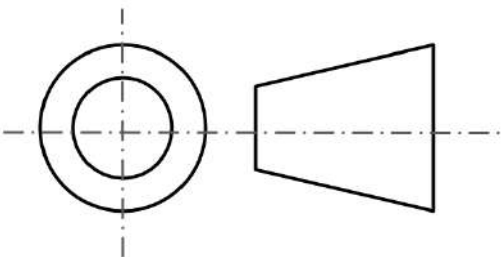
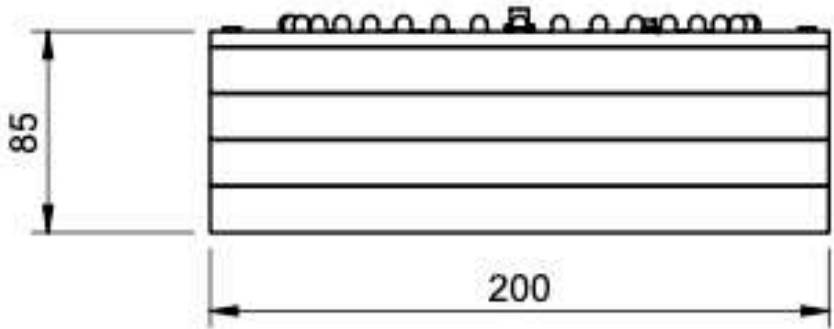
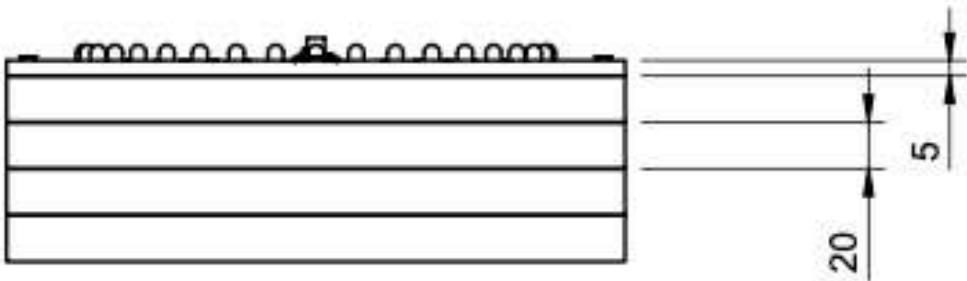
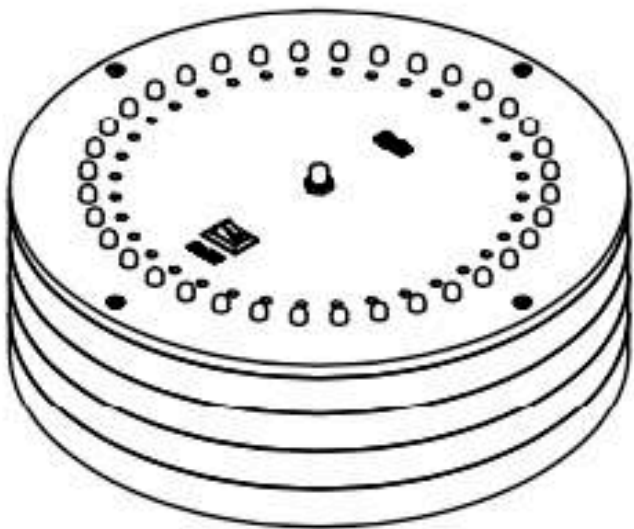
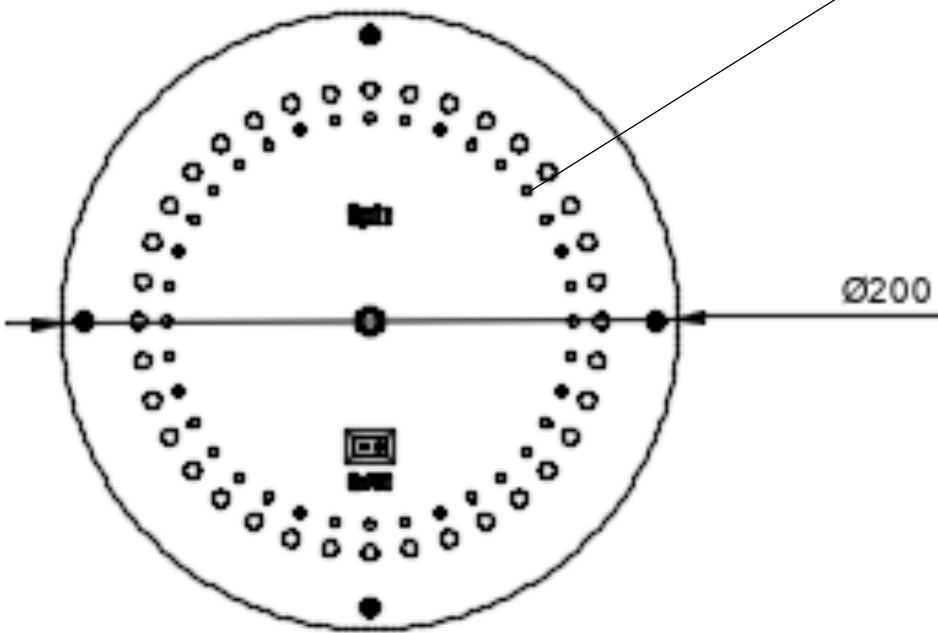
C5



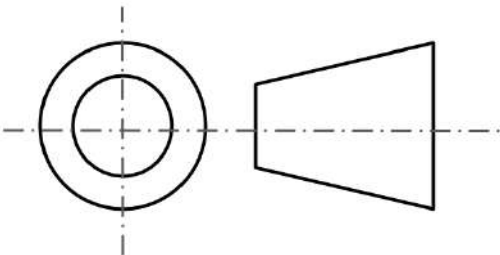
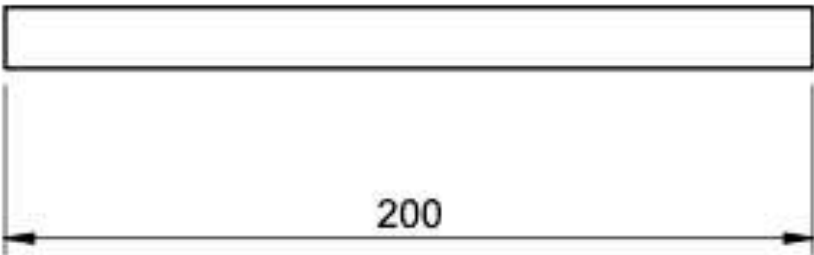
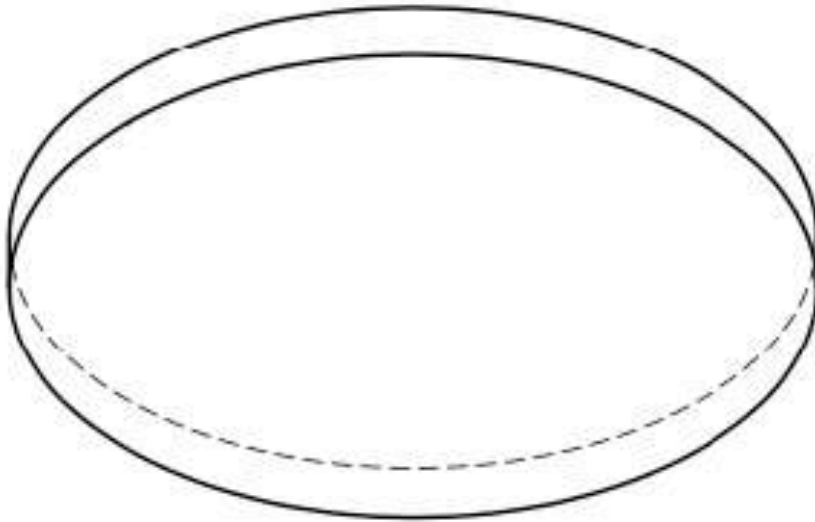
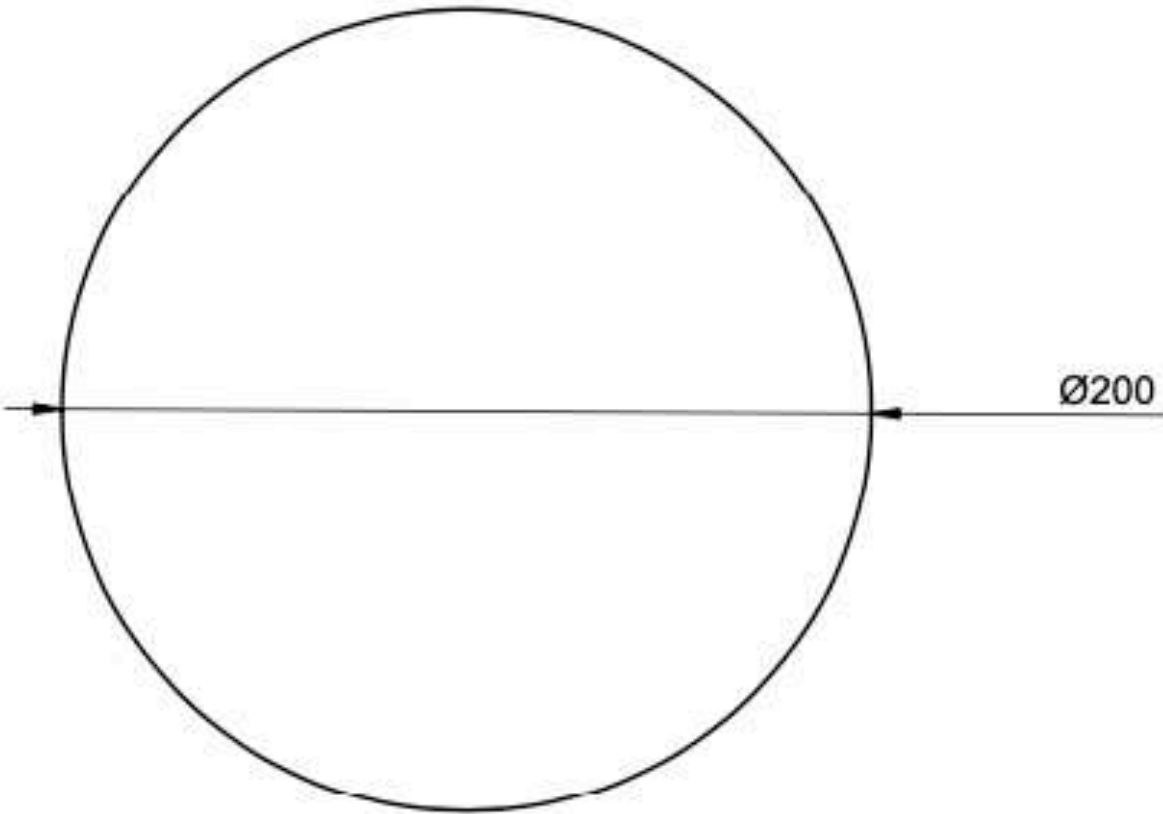
Note: each white dot on the inside ring represents a number from 1 to 36, with the dot above the text “spin” being 1 and the number increasing as you go clockwise



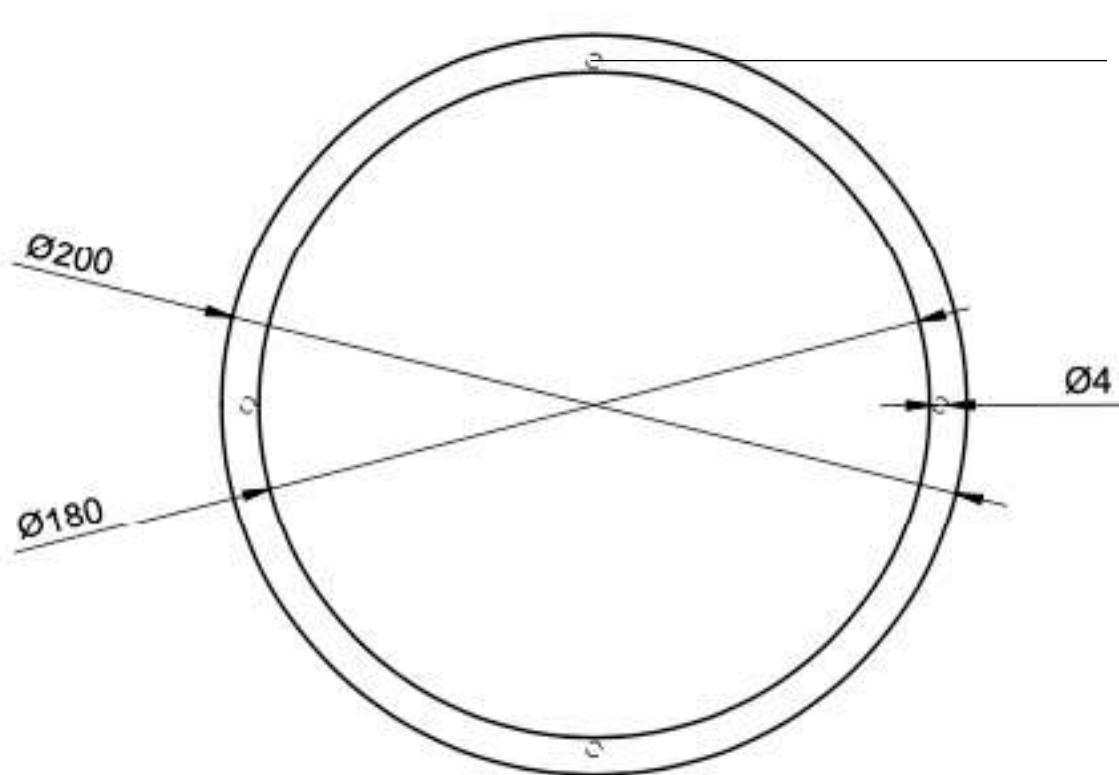
Note: each dot on the inside ring represents a number from 1 to 36, with the dot above the text “spin” being 1 and the number increasing as you go clockwise



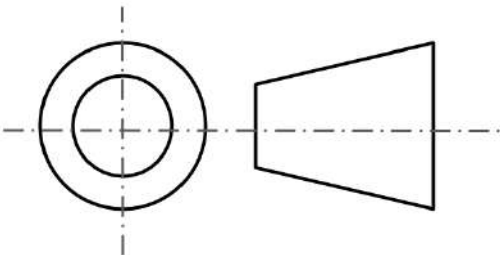
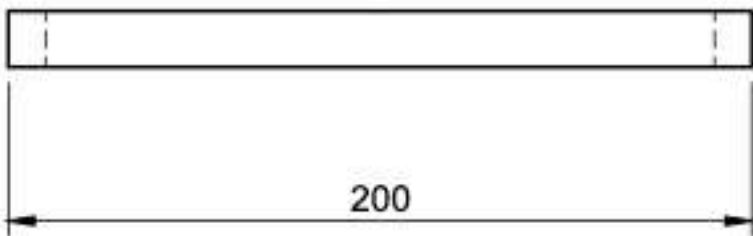
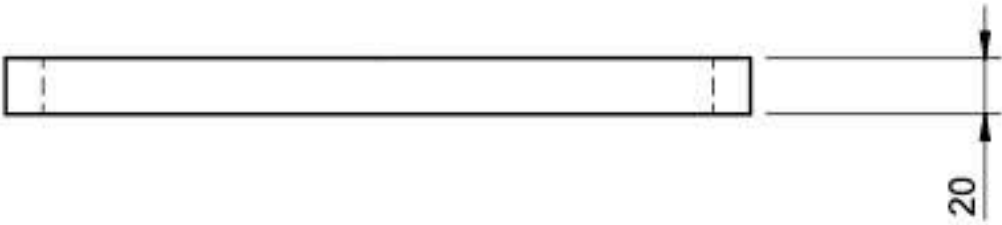
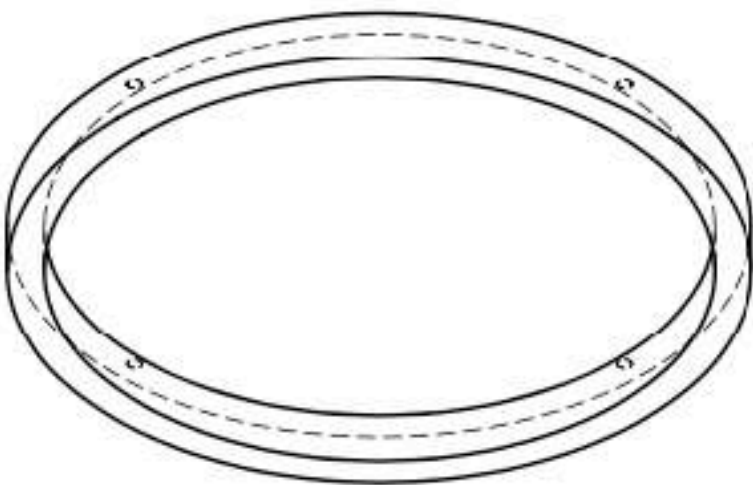
Component name: Electronic roulette wheel (full product)	Materials: Kapok and black acrylic
Units:mm	Quantity: 1



Component name: Housing base	Materials: Kapok
Units:mm	Quantity: 1

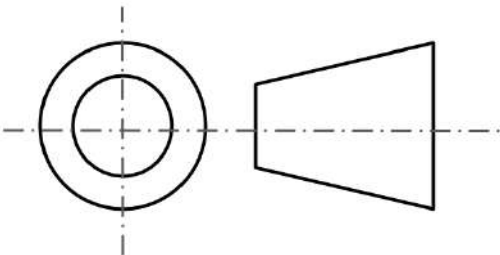
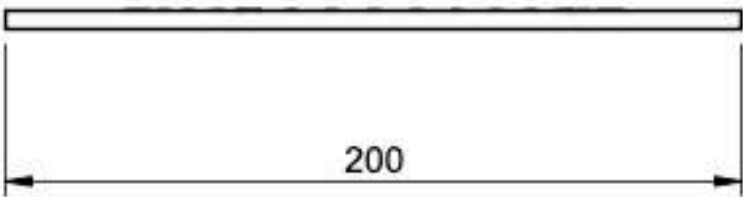
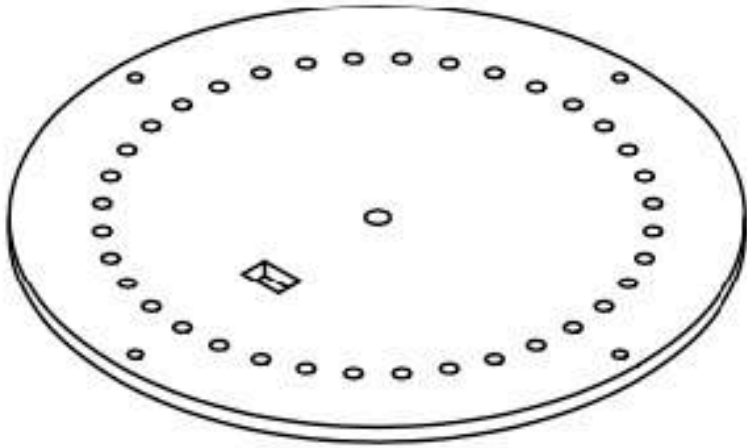
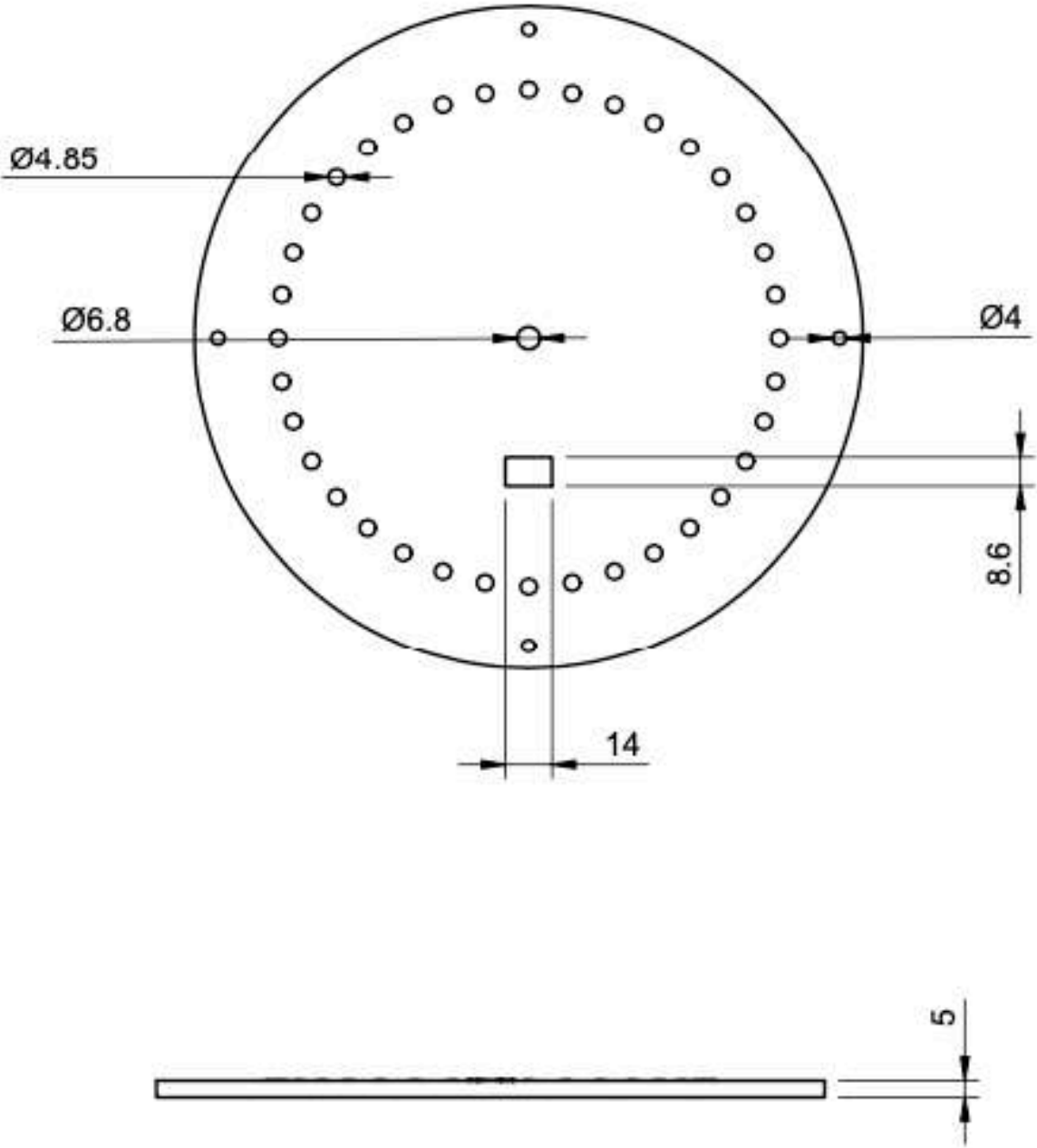


Note: 4mm holes are only for the top layer of the housing because of screws, other layers don't have holes

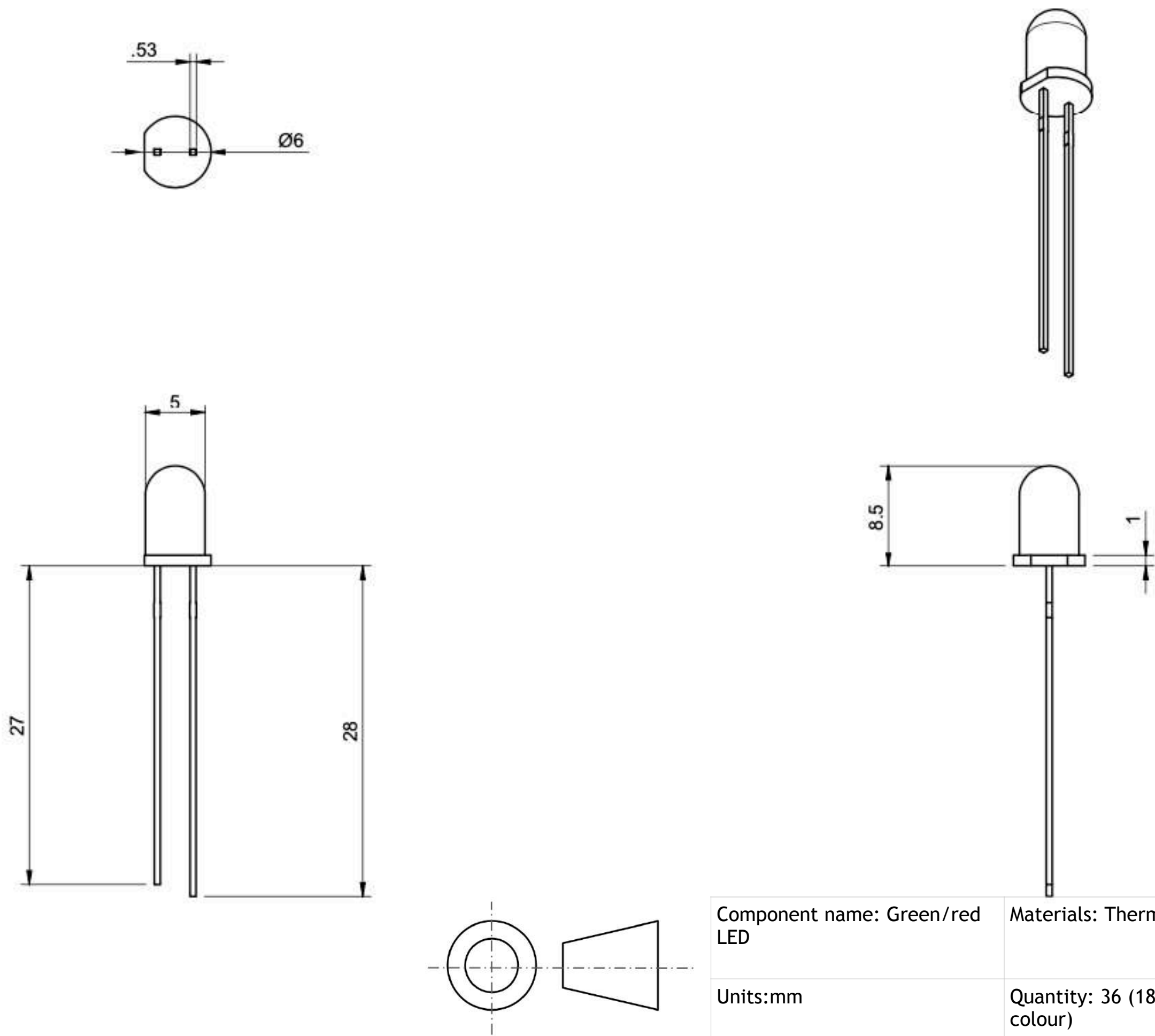


Component name: Kapok housing body	Materials: Kapok
Units:mm	Quantity: 3

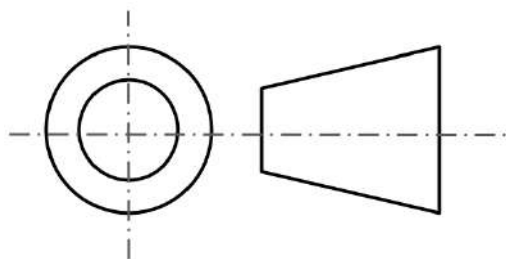
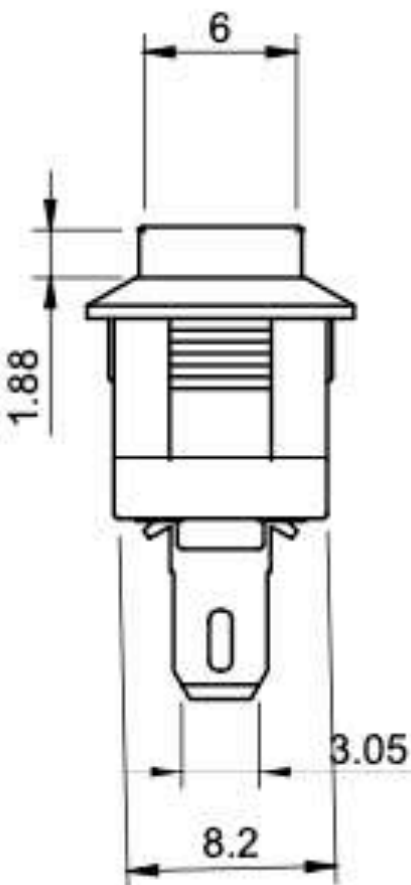
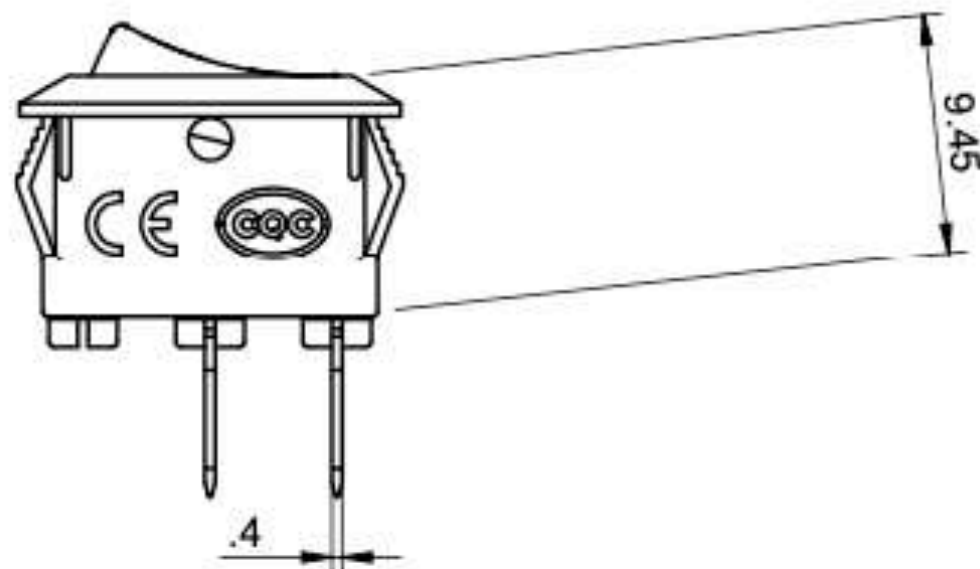
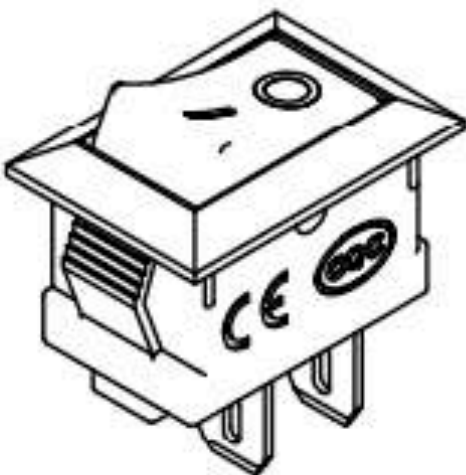
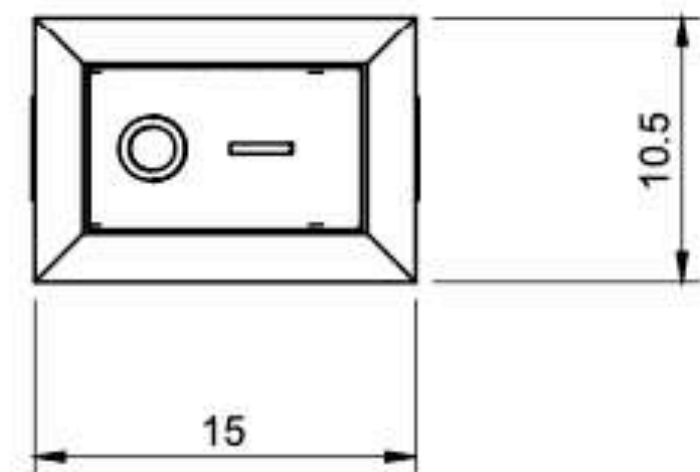
Note: User interface text is not shown on orthographic



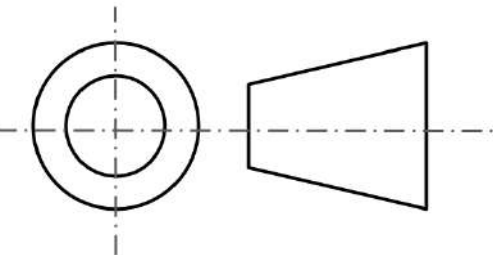
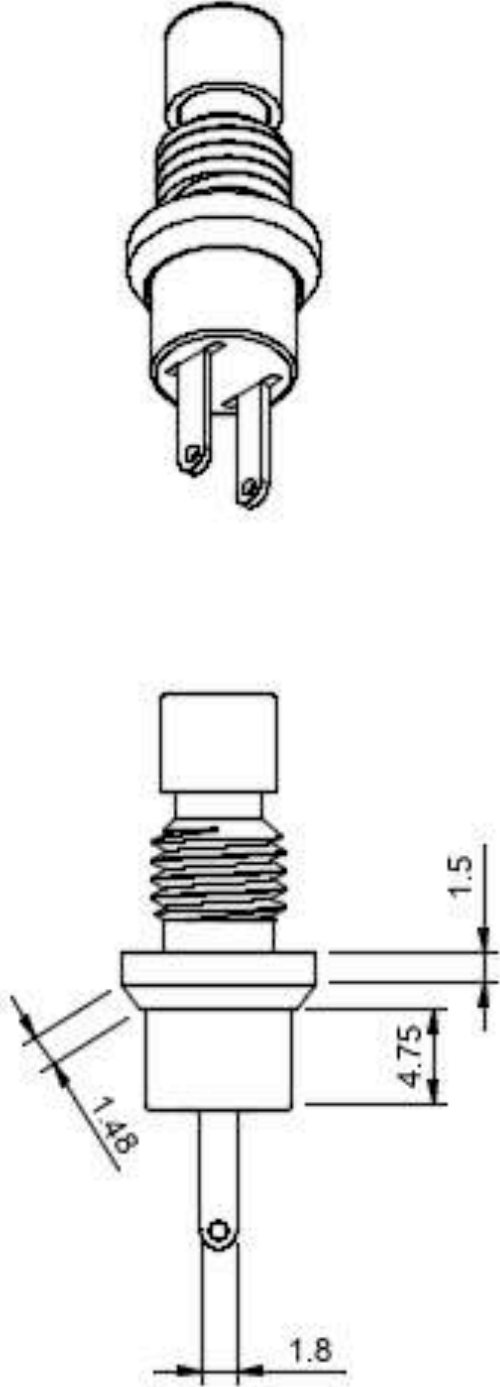
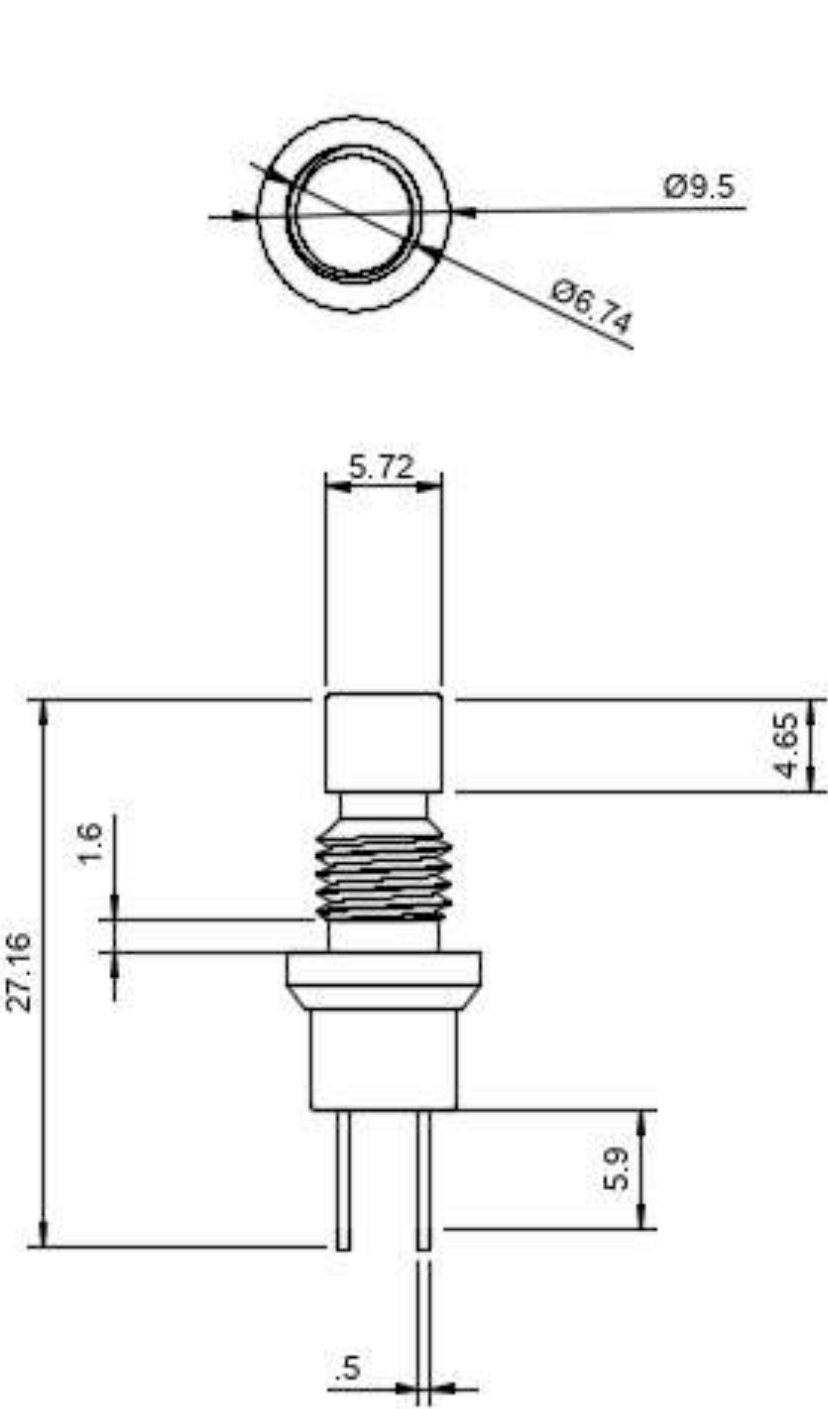
Component name: User interface	Materials: Black acrylic
Units:mm	Quantity: 1



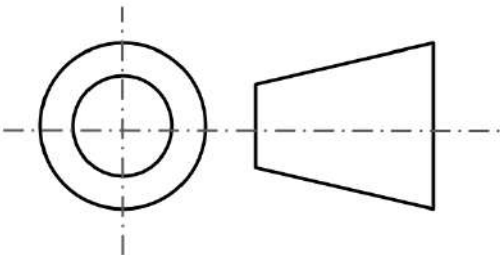
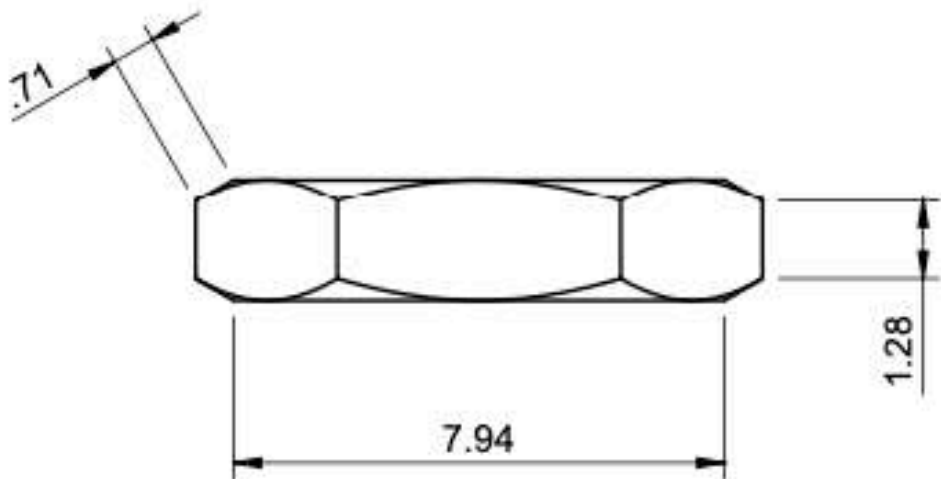
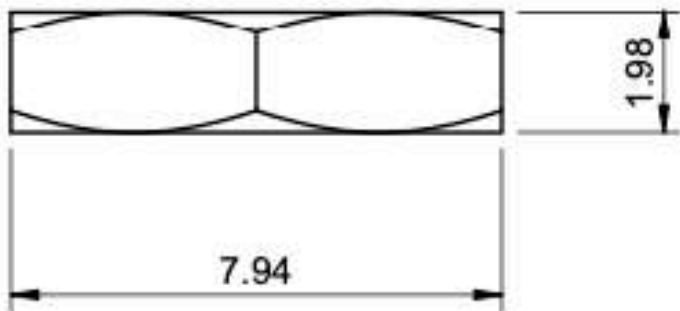
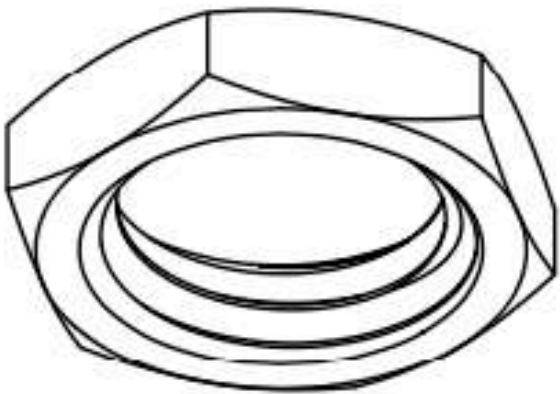
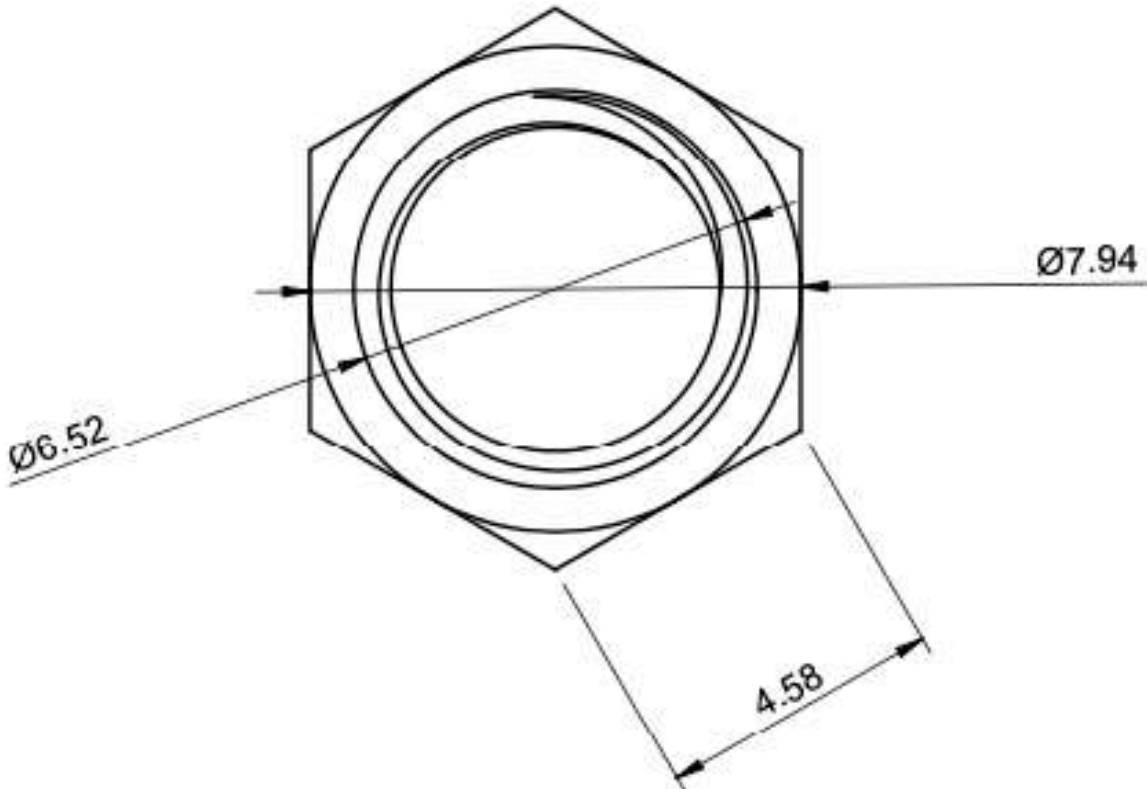
Component name: Green/red LED	Materials: Thermoplastic
Units:mm	Quantity: 36 (18 of each colour)



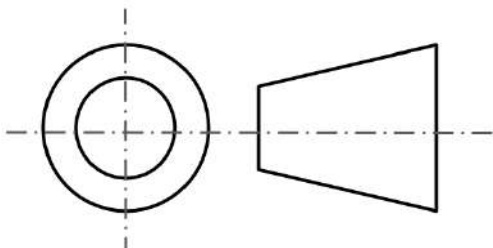
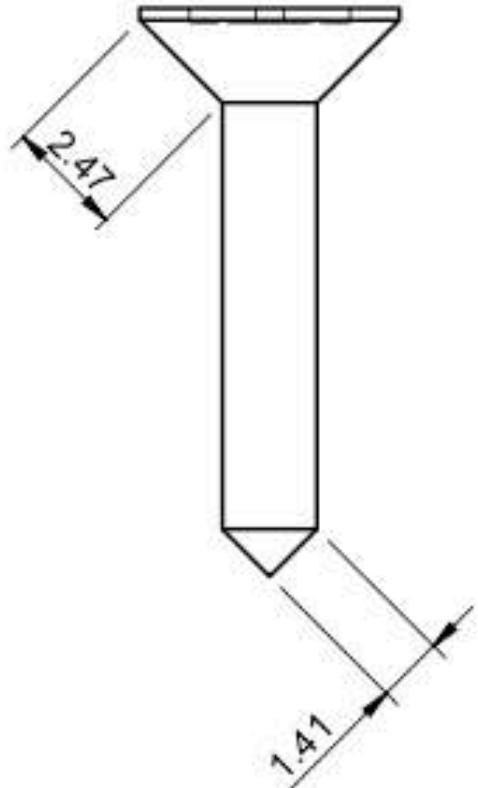
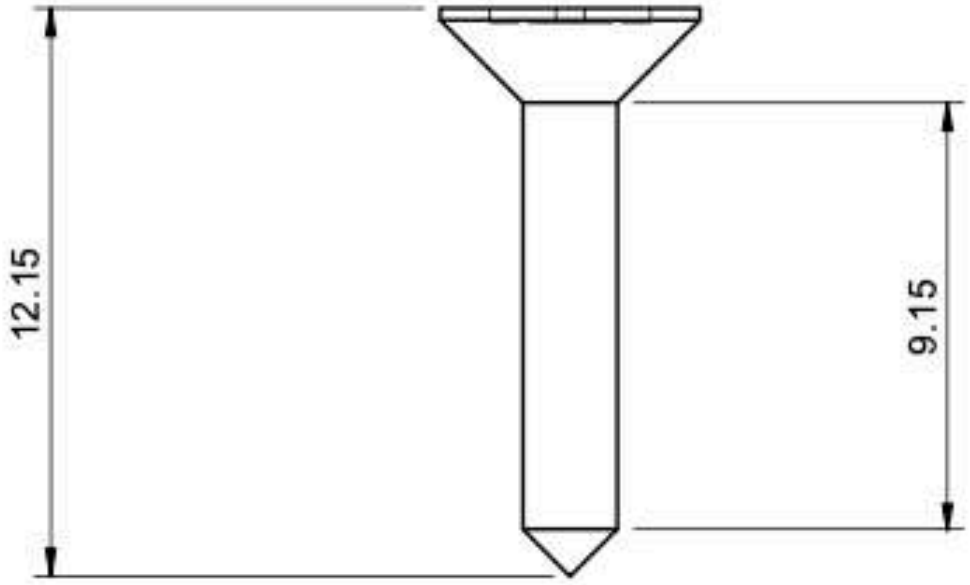
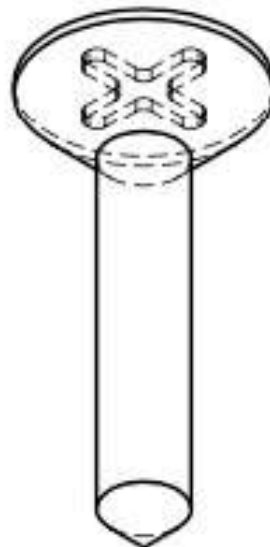
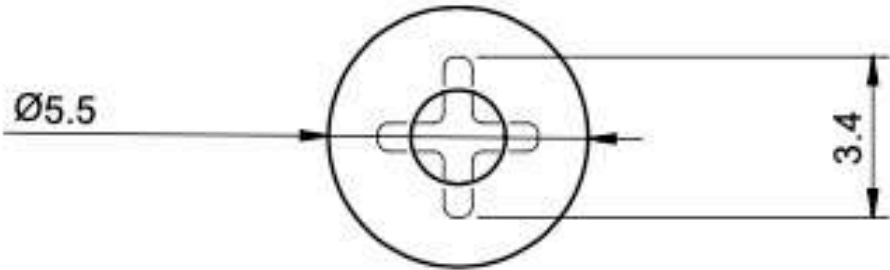
Component name: Rocker switch	Materials: Thermoplastic
Units:mm	Quantity: 1



Component name: Push to make (PTM) switch	Materials: Thermoplastic
Units:mm	Quantity: 1



Component name: Hex nut	Materials: Steel
Units:mm	Quantity: 1



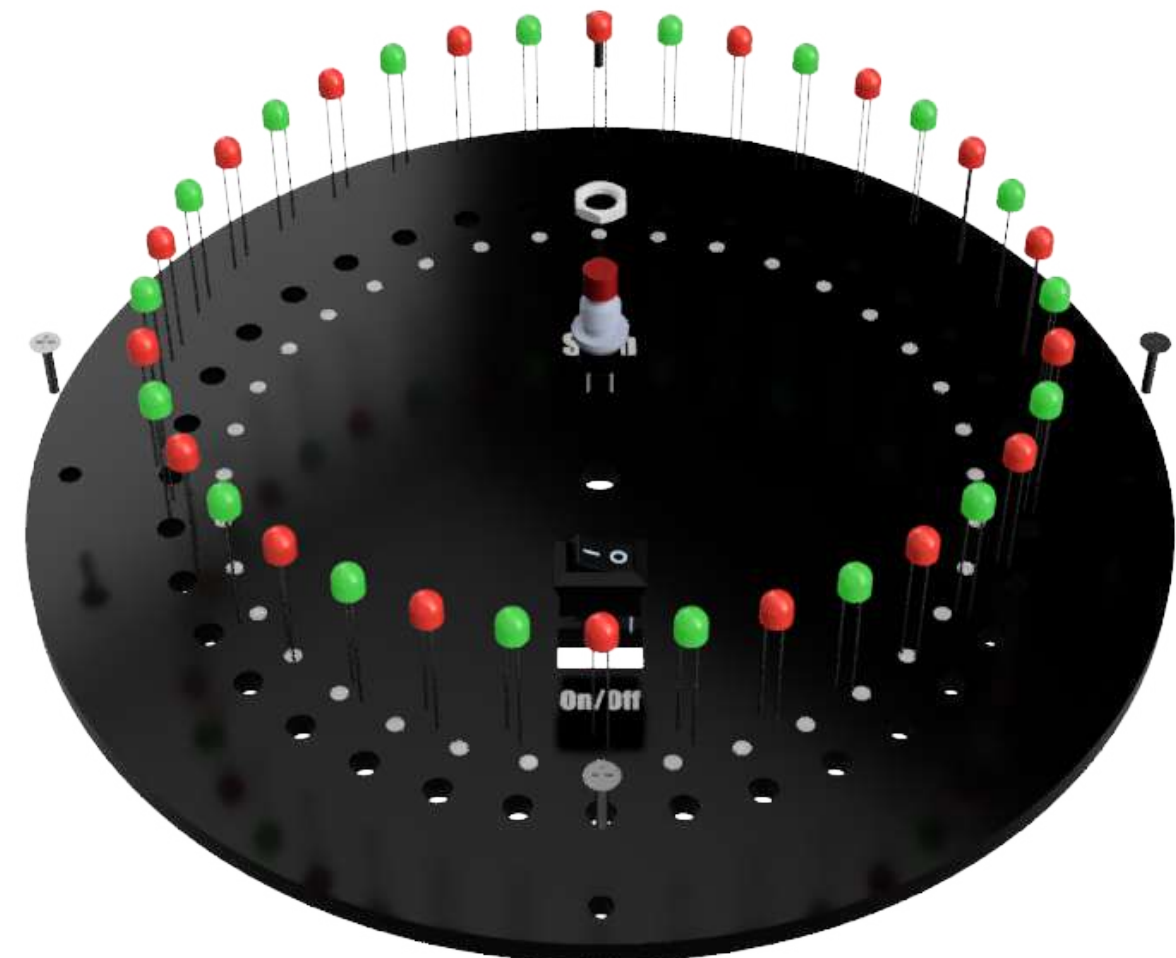
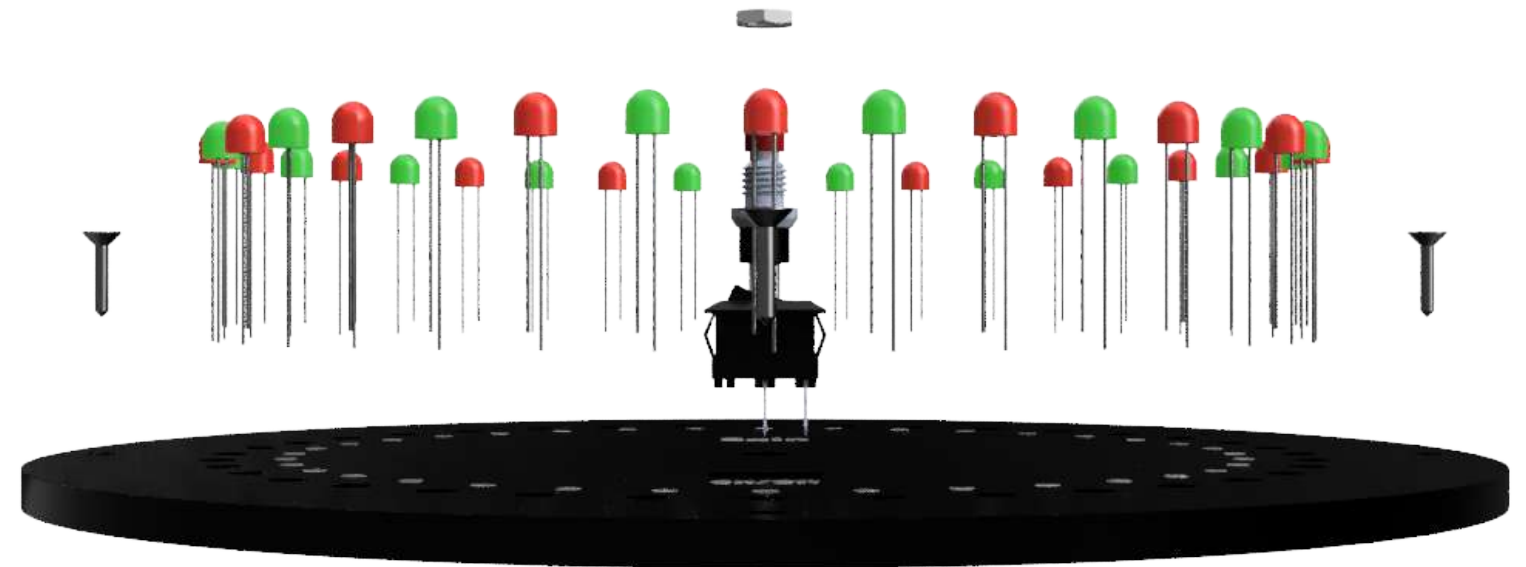
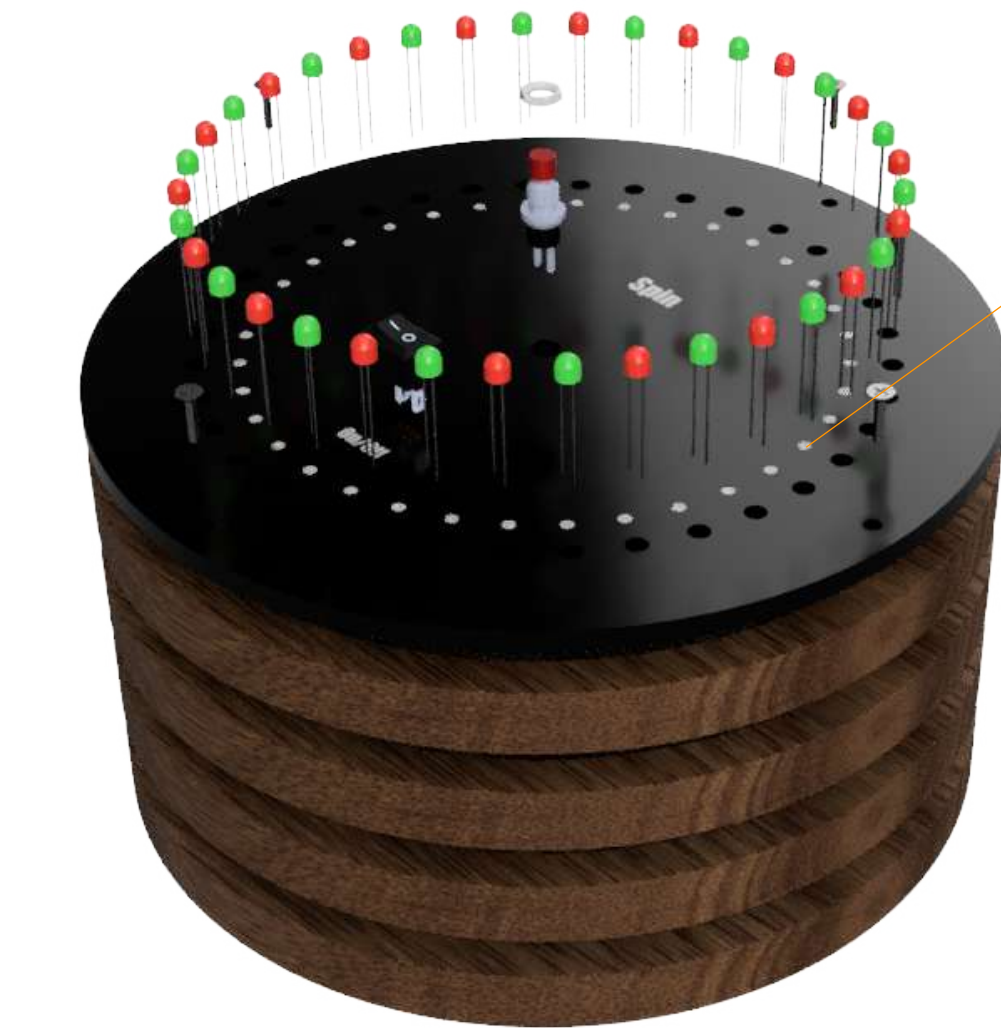
Component name: Wood screw	Materials: Steel
Units:mm	Quantity: 4

Rendered Exploded View

The following page presents the rendered exploded view of the product, rendered and made in fusion 360

C5

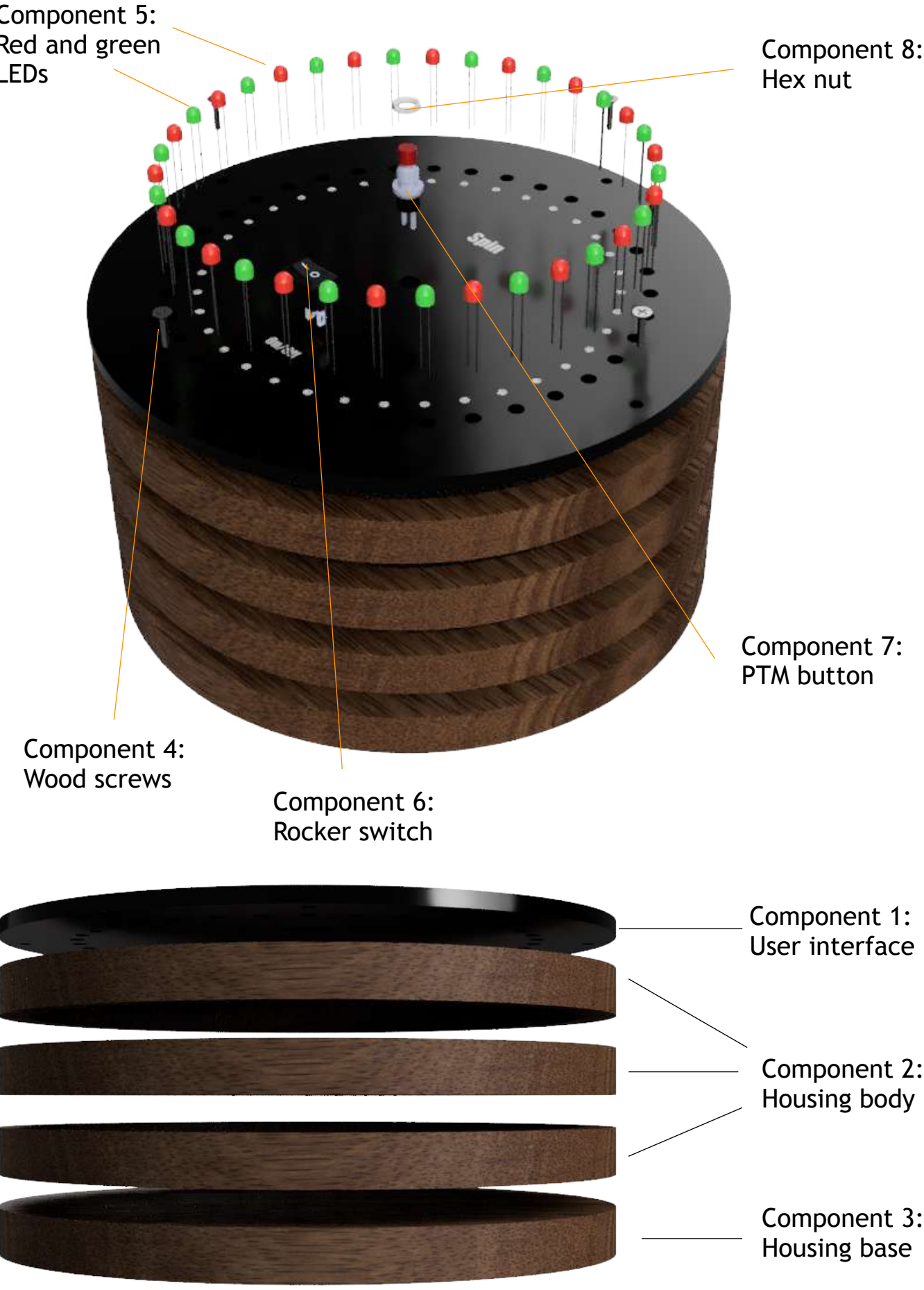
Note: each white dot on the inside ring represents a number from 1 to 36, with the dot above the text “spin” being 1 and the number increasing as you go clockwise



Rendered Exploded View and Cutting List

The following page shows a rendered exploded view of the product made on fusion 360 and includes a short list of the materials and their properties used in the design

C5



Component	Material description	Quantity
Component 1: User interface	Black acrylic • Lightweight • Good impact resistance • Durable	Quantity: 1
Component 2: Housing body	Kapok • Hard and strong, good strength to weight ratio • Moisture resistant • Resilient and withstands shocks	Quantity: 3
Component 3: Housing base	Kapok • Hard and strong, good strength to weight ratio • Moisture resistant • Resilient and withstands shocks	Quantity: 1
Component 4: Wood screws	Steel • Tough • Strong • Hard	Quantity: 4
Component 5: Red and green LEDs	Thermoplastic • Durable • Lightweight	Quantity: 36 (18 of each colour)
Component 6: Rocker switch	Thermoplastic • Durable • Lightweight	Quantity: 1
Component 7: PTM button	Thermoplastic • Durable • Lightweight	Quantity: 1
Component 8: Hex nut	Steel • Tough • Strong • Hard	Quantity: 1

Preparation of components

Materials: Circuit and circuit components

Processes: Insert circuit components into circuit board



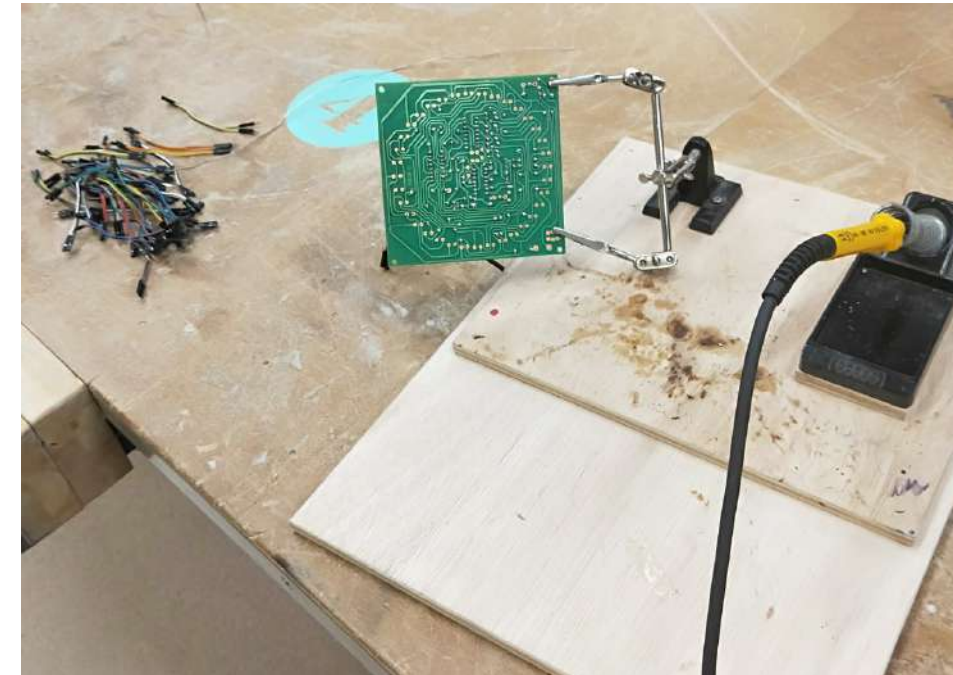
Installation and soldering of components

Materials: Circuit board with components

Tools: Soldering iron, soldering station, solder

Processes: **Solder** components to circuit board using soldering iron and station

Safety: Don't burn yourself



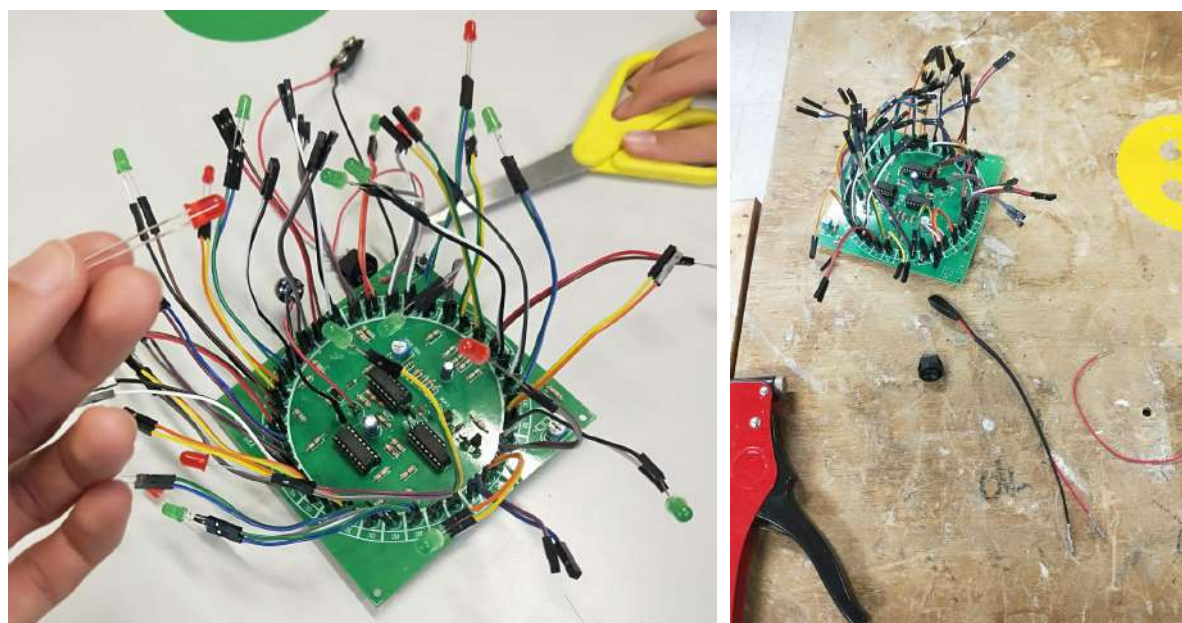
Stripping and soldering wires

Materials: Wires, battery clip, circuit

Tools: Wire cutters, soldering iron, soldering station, solder

Processes: **Strip** wires with wire stripper and **solder** external wires to circuit board for external components

Safety: Don't burn yourself

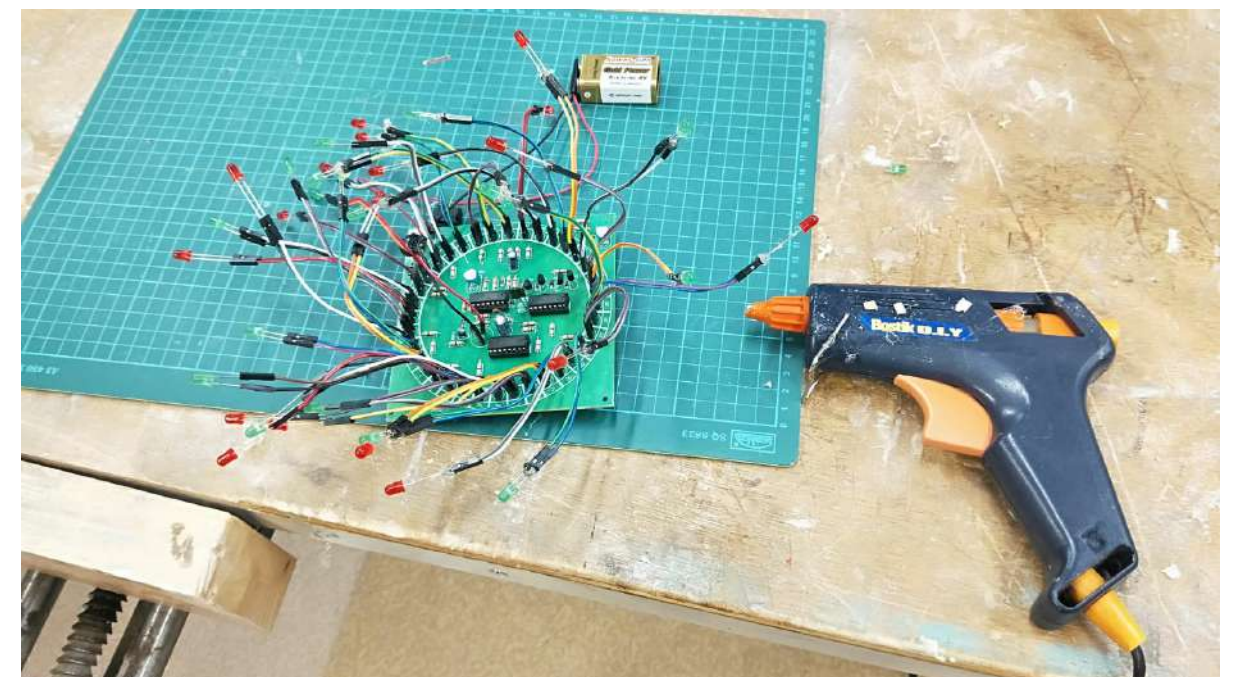


Installing LEDs

Materials: LEDs, circuit board

Tools: Hot glue gun

Processes: **Insert** LEDs to male female wires and glue with hot glue gun



Production Process

Housing

C6

Preparation of materials

Materials: Kapok
Tools: Ruler
Processes: **Measure** dimensions of kapok using ruler



Cutting

Materials: Kapok
Tools: CNC milling machine
Processes: **Cut** out Kapok plank using CNC machine
Safety: Get supervision or let someone more skilled perform the cutting



Joining

Materials: Kapok cut housing
Tools: Resin, clamp
Processes: **Apply** resin on intersections of housing and use clamp to hold housing together to dry



Shaping

Materials: Kapok housing
Tools: Disk sander
Processes: **Sand** down the edges of the housing with the disk sander
Safety: Be careful with fingers



Finishing

Materials: Kapok housing
Tools: Osmo oil, drying rack
Processes: **Apply** osmo oil onto surface of wood using polishing cloth, then **leave** on drying rack to dry



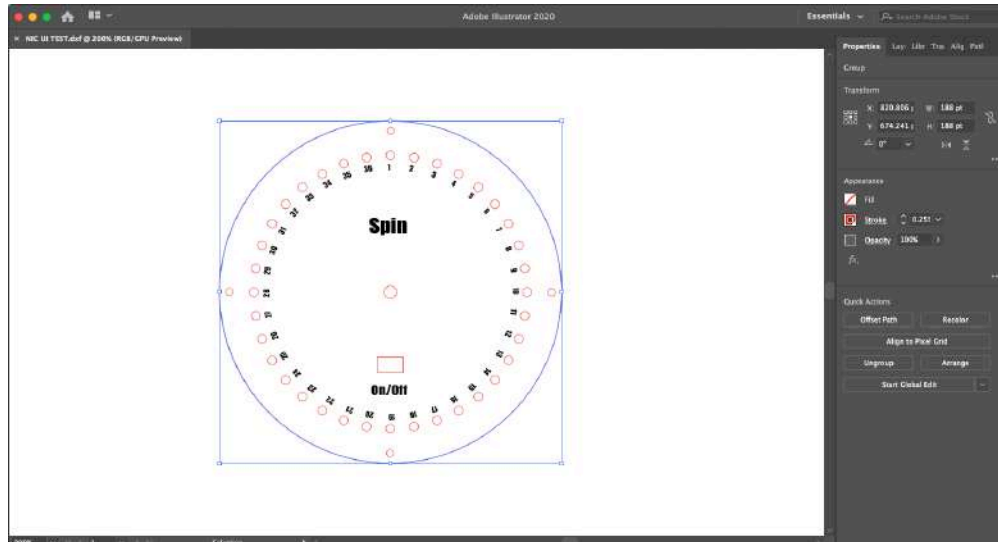
Production Process

Laser cutting

Materials: Black matte acrylic

Tools: Laser cutter

Processes: Send DXF file to laser cutter and cut out user interface



User Interface and Finishing

C6

Colouring in text

Materials: User interface

Tools: White marker

Processes: Colour over text on user interface with white marker



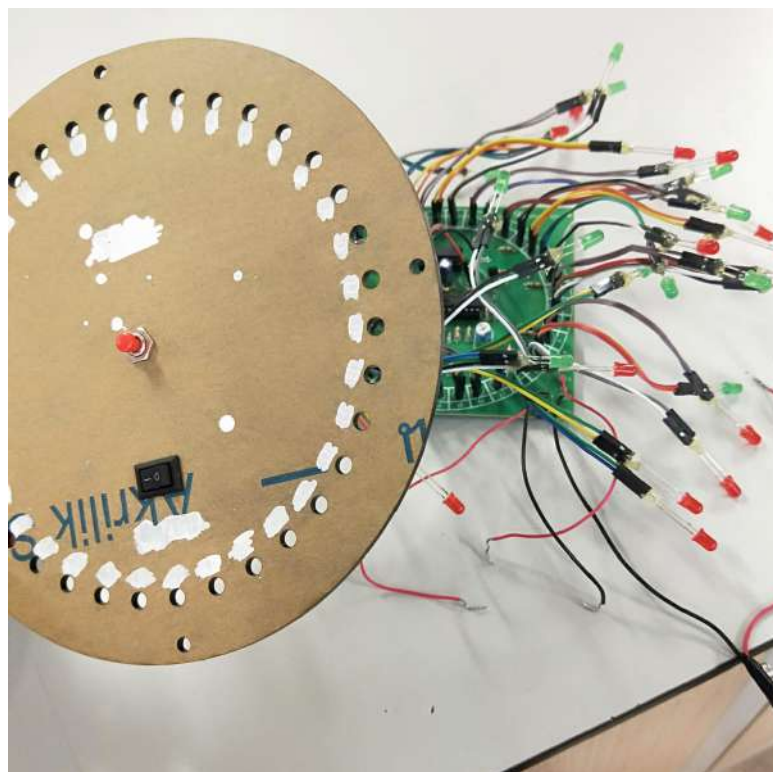
Joining user interface and circuit

Materials: User interface, circuit

Tools: Solder, soldering iron, soldering station

Processes: Insert LEDs and circuit components through user interface then solder

Safety: Don't burn yourself



Assembly

Materials: User interface + circuit, housing

Tools: Screwdriver

Processes: Insert circuit into housing and use user interface to cover housing. Then peel off user interface protector and screw in screws with screwdriver



Justification of Modifications

Whilst making my product, I have decided to make some **modifications** along the way to either make my design more **aesthetically pleasing** or to improve the **function**. Here are the modifications I made.



Original design

Modification 1:
Using matte black acrylic instead of regular glossy black acrylic

Justification:
This modification creates a more complete **unique appearance** of the design. Instead of using the usual glossy black, I felt as though using matte black would allow the bright LED colours to **stand out** more as the surface was **less reflective**

Modification 2:
Glueing LEDs to user interface

Justification:
This modification was made as the LEDs were all slightly **different in size**, causing some LEDs to be loose. Because of this, I decided to **glue** the LEDs to the user interface to **secure** them and to ensure that the LEDs don't **detach**



Final design

Modification 3:
Slightly shrinking the size of the user interface

Justification:
This is due to the **sanding** of the housing. Sanding the housing has made the edges **smoother** but has slightly **shrunk** the size. Testing with cardboard has shown that the regular 200mm user interface was **slightly too large** for the housing. The size of the user interface was **reduced to 188mm** to match the size of the housing



Final product in intended environment



Testing and Evaluation in Intended Environment

Turquoise = space available

C7

The following page will test and **evaluate** photos of the product in the intended environment. Due to my client rearranging the furniture at their home, the shelf and playing space has been **changed**. The new dimensions will be provided below.



Storage

Specification: Product must be smaller than 34 x 37.5 cm and shorter than 40 cm

The product clearly has a lot of **space** and definitely follows the specification. There is a lot of unused space that the client can use for **other purposes** and will ensure that the product won't fall off the shelf. The product is also easily **accessible** for the clients and can be taken anytime, with the desk being directly under the shelf (shown in middle left photo in the page before). However, one small downside is that the product **doesn't blend** in with the shelf.

Positives:

- Lots of unused space
- Product won't fall off shelf
- Product is easily accessible by clients

Negatives:

- Product doesn't blend in well with the shelf

Product on desk (playing space)

Specification: Product must have a modern look and must be smooth and sleek

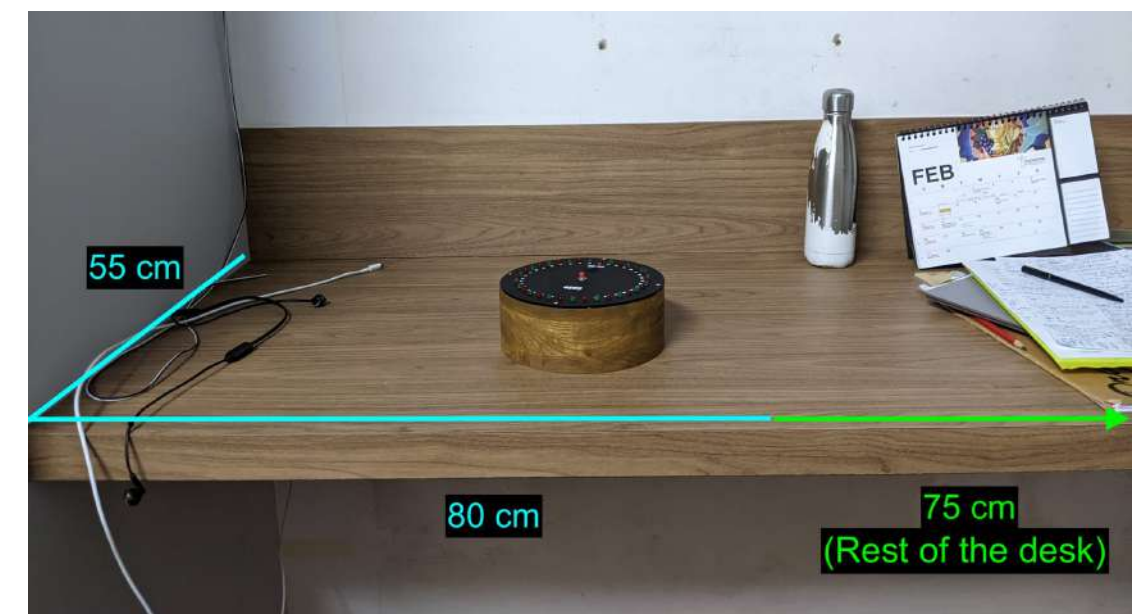
The product has a modern look that **blends** in very well with the desk. The dark kapok wood of the product **matches** very well with the dark wood of the desk. The amount of **space** allocated for playing may **fluctuate** due to the amount of objects on the desk. The desk also provides a large amount of **space** for the clients to use the product. One downside to the desk is that the clients **cannot be face to face** when using the product.

Positives:

- Lots of extra space for product on desk
- Modern look that blends with desk (wood)

Negatives:

- Clients cannot be face to face



Product in use

Specification: Product must be a game that allows a form of betting between clients and has a large number of possible outcomes to be entertaining

The product is definitely a device that **allows betting** between the clients and has 36 possible outcomes. The game should be interesting to play with factors such as **sound** and the **cycling of LEDs** to build up **suspense** and **entertain** the clients. One downside is that there is only **one mode/setting**, which is having the roulette wheel cycle after pressing the push button with no other option. However to counter this, the clients are open to use their **imagination** and **creativity** to create other games with the roulette wheel as there are 36 possible outcomes.

Positives:

- The product allows betting
- 36 possible outcomes
- Has high potential to be entertaining for clients
- Allows clients to be creative

Negatives:

- Only one mode/setting on roulette wheel

Testing and Evaluation Against the Specifications

The following page will test and evaluate my product against the specifications

0 = not met expectations
1 = met expectations
2 = exceeds expectations

C7

Specifications	How this was tested	Justification	Score (self)	Client evaluation (interviewed Satvik)	Self evaluation	Score (Final)
Function - Solution must be a game that allows a form of betting between the clients - Product needs to have large number of possible outcomes to be entertaining		- Product allows betting between 2 clients - Product has 36 possible outcomes and entertains clients	2	"This product seems to be a good solution to our problem and definitely won't get boring too soon. Having 36 possible outcomes will definitely make things interesting for us. One small issue is that once in a while, a second LED turns on dimly next to a lit LED when the cycling stops"	I am quite pleased with the product and I feel it definitely serves as a great solution. I feel as though I chose the correct circuit to make, however, the issue with the second LED that my clients mentioned is quite unfortunate. Even after troubleshooting I could not solve this issue and so chose to ignore the problem as it shouldn't affect the product too much	1
Aesthetics - Product must have a modern feel - Product must be smooth and sleek - Product must not look dull and plain		- Product placed on the modern shelf presents a modern feel with it's dark colours but doesn't match well with shelf - Product has smooth edges as it has been sanded - Product has natural wood patterns and a detailed user interface with dark colours	2	"The product definitely has a modern vibe with a cool detailed user interface that my brother (Varun) and I really like. The wooden smooth texture and patterns are also pretty cool. The product also matches with my desk, however not as much with the shelf"	I definitely tried my best to keep the colours more dark and modern such as the wood and the acrylic. I am happy with my clients' evaluation as I feel they are satisfied	2
Materials - Material must be durable and last over a long period of time - Material must have a modern style - Material must be hard		- Acrylic and kapok wood are durable - As said above, product presents modern style with dark kapok wood and black acrylic - Acrylic and kapok wood are both hard impact resistant materials	2	"The product feels very sturdy and feels like it could last a long time . I have no doubt that the product will be able to survive a few bumps or falls "	I am proud of my choices of materials and know that these were one of the best options I could've made. Kapok and acrylic are both hard and durable materials which is what my clients need	2
Cost - Materials and circuit must be below \$80 SGD	Circuit - \$30 SGD Housing - \$20 SGD Acrylic - \$1.50 SGD Circuit components - \$4 SGD	- Product is below \$80 SGD and totals to \$55.50 SGD	2	"This is great as it falls in our price range and is super affordable"	I am satisfied with the price of the overall product and don't think I can change much	2
Size - Product must be smaller than 34 x 37.5 cm - Product must be shorter than 40cm - Product must not be too heavy. Lighter than 1 kg		- Using a ruler to measure, the product has a diameter of just under 20cm and a height under 10cm - The product weighs 883 grams	2	"The product fits well on our shelf which is what matters most as that it where we will be storing the product when not in use"	I am happy with the size of the product as I felt as it is the smallest I could've made with the size of my circuit	2
Safety - Product must not have sharp edges - Product must not be too heavy nor too light - Product should not have loose parts or loose circuits sticking out		- There are no sharp edges and only curved edges - The product weighs 883 grams - No lose parts that could cause harm to the client are sticking out	1 Total: 11/12	"The product doesn't seem like it could cause any harm to us and has a decent weight for its size. The product also has no sharp edges which is something we are glad about"	I feel I have done a pretty good job with the finishing, leaving no harm for my clients. After seeing my clients' feedback I am glad they think so as well and are certain that the product is completely safe	2 Total: 11/12 39

Proposal for Future Developments

C7

In this page, I will state my proposals for future developments by evaluating current positives and negatives

Manufacturing and construction methods

Positives:

- Construction worked well overall and ended with a complete product

Negatives:

- Overall production process is long and takes a long time

Proposal for future developments:

- No proposal, however if possible try to skip or shorten steps in production process

Finishing

Positives:

- Looks finished with a dark appearance
- Smoothened sides

Negatives:

- Osmo oil may have to be reapplied after a while

Proposal for future developments:

- Use a finishing solely for durability so osmo oil won't have to be reapplied for example Tried and True Original Wood Finish

Joining methods

Positives:

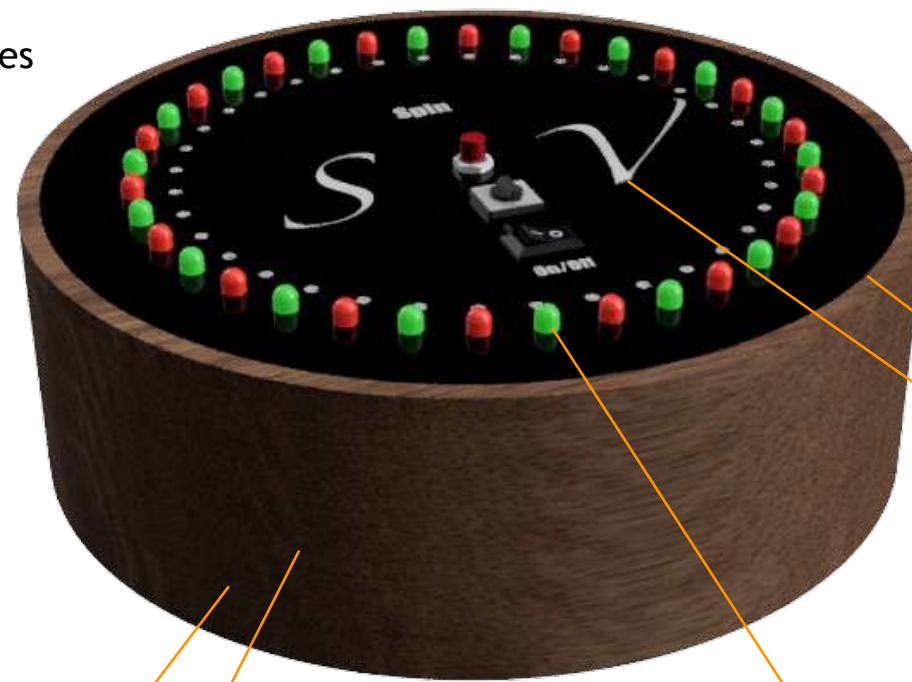
- Joined tightly with no loose parts

Negatives:

- Screws may look ugly

Proposal for future developments:

- Have the top sealed off completely with resin and have a battery case at the bottom that can be opened with a cover



Shape and form

Positives:

- Minimal sharp edges
- Aesthetically pleasing circular shape
- Strong base

Negatives:

- May appear large and bulky
- Edge of user interface is exposed
- Wires are pressed with little space inside housing

Proposal for future developments:

- Use an indented user interface
- Use shorter male female wires
- Add logo for decorative purposes and to show brothers' possession

Function

Positives:

- Solves the clients' problem
- Product has a large number of outcomes

Negatives:

- Not as light and portable as desired
- Product may be repetitive over a long period of time
- Two LEDs lighting issue

Proposal for future developments:

- Use a more complex circuit with multiple modes, for example have a random mode that causes LEDs to light up in a random order instead of clockwise

Materials

Positives:

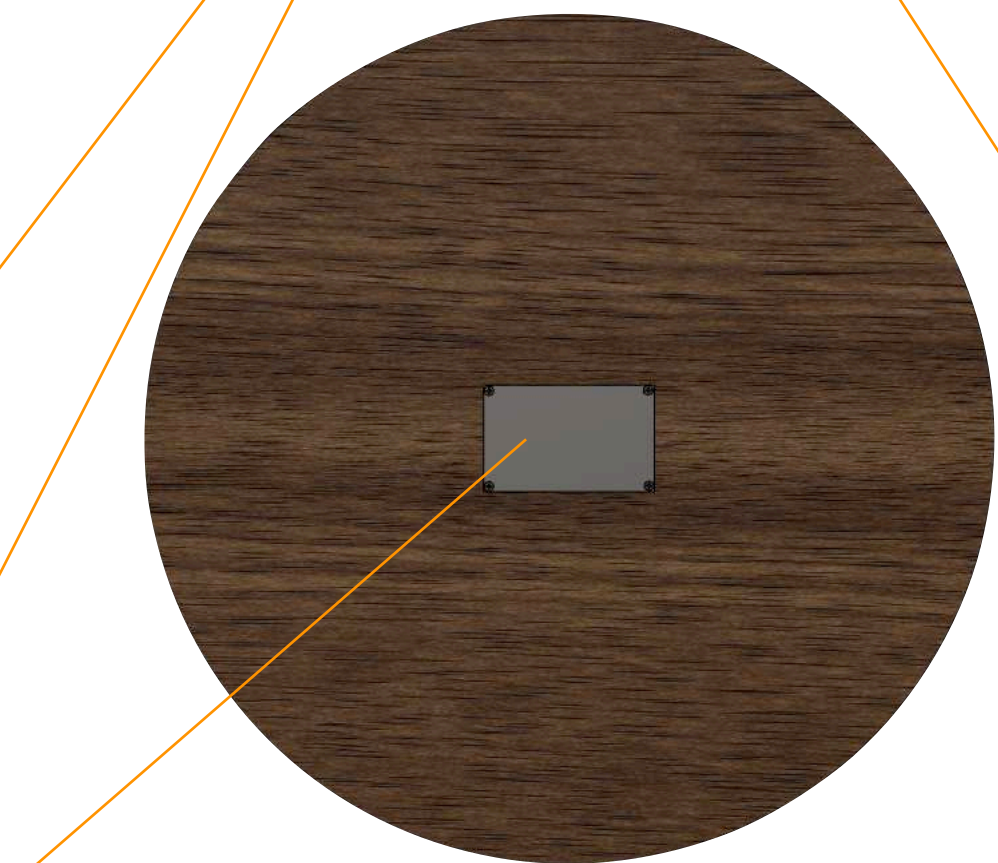
- Strong, hard and durable materials used
- Modern style and colour

Negatives:

- Moderately heavy product
- Susceptible to scratches

Proposal for future developments:

- No proposal, in my opinion, the materials used are near perfect



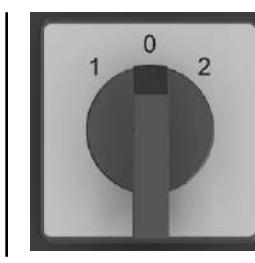
Metal flap to protect battery



8 cm

5 cm

Rotary switch for multiple modes (complex circuit)



12 cm

12 cm

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Component Credits	
<i>Free CAD designs, Files & 3D models: The grabcad community library</i>. Free CAD Designs, Files & 3D Models The GrabCAD Community Library. Rotary switch - https://grabcad.com/library/electrical-part-power-switch-660v-lth-125a-on-off-on-3-positions-1 PTM button - https://grabcad.com/library/r13-24-push-button-switch-1 Rocker switch - https://grabcad.com/library/small-power-switches-rocker-type-10-5x15mm-1 LED - https://grabcad.com/library/standard-5-mm-0-24-in-led-1	27-31
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